

Implicit Representations of 3D Geometry: A Comprehensive Survey, Benchmark, and Future Directions

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Implicit representations encode 3D geometry via field functions and associated geometric constraints that jointly characterize the target shape, naturally supporting gradient-based optimization and offering topological flexibility. Existing surveys adopt either a task-oriented or a network-oriented perspective, leaving the relationships among field design, surface extraction, and downstream deployment insufficiently systematized. This gap makes it difficult for researchers entering geometry modeling to form a structured understanding of implicit representations and their practical trade-offs. This survey provides a unified, representation-centric treatment of implicit 3D geometry, tracing the complete pipeline from field formulation to explicit surface extraction and downstream applications. We first formalize major implicit field families, including occupancy fields, signed distance fields, and unsigned distance fields, and their vector-, ray-, segment-, and medial-variant extensions, under a common abstraction defined by query domain, value domain, and geometry condition. This abstraction clarifies how different fields trade off expressiveness, regularity, and compatibility with downstream tasks. We then review iso-surface extraction algorithms that convert implicit fields to explicit meshes, followed by two representative applications: surface reconstruction and 3D generation. To support a fair and reproducible evaluation, we further introduce a unified benchmarking platform with a standardized interface covering 19 representative methods, enabling consistent evaluation of existing approaches and straightforward integration of future ones. The benchmark reveals empirical findings that, together with our systematic analysis, inform a discussion of open challenges such as non-manifold topology recovery, faithful reconstruction of open and thin-sheet geometry, and locally controllable neural representations. Finally, we discuss challenges and emerging directions, and outline a few future paths forward.

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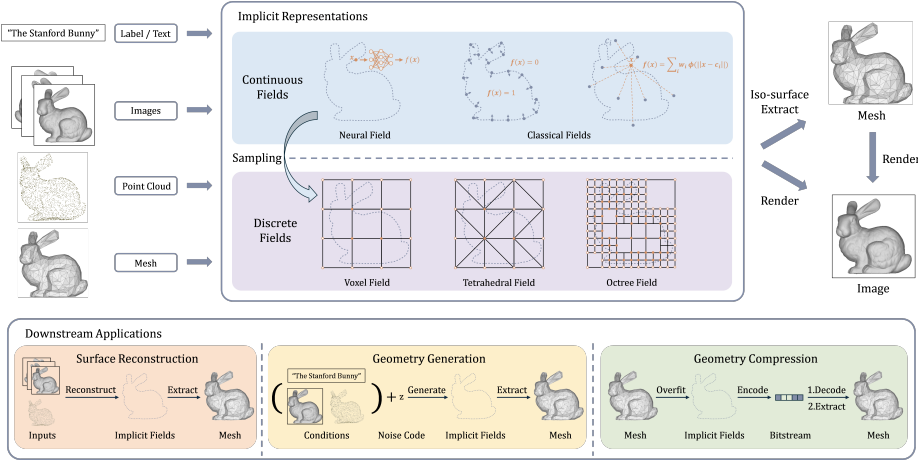


Fig. 1. Overview of the implicit representations.

1 INTRODUCTION

The rapid proliferation of digital twin technologies [51, 213], autonomous robotics [13, 89], immersive communication [162, 90], and spatial computing [147, 81] has propelled 3D geometry data to the forefront of modern computer science. As the bridge between physical reality and virtual simulation, 3D geometry data is indispensable to high-stakes applications ranging from medical imaging [148] and heritage preservation [92] to the generative pipelines of world models [63]. The choice of geometric representation governs the fidelity, efficiency, and flexibility of all downstream algorithms. Explicit representations specify the surface directly. The most common are polygon meshes, the standard output across geometry processing pipelines, which offer efficient rendering and manipulation. Other examples include point clouds, valued for flexible acquisition, and parametric surfaces such as NURBS [170], which provide mathematically precise, smooth definitions central to computer-aided design. However, the geometric detail of meshes and point clouds is inherently limited by the number of primitives, while parametric surfaces require tessellation for rendering and are difficult to fit to arbitrary topologies. Meshes in particular pose an additional challenge for modern learning-based pipelines: although their vertex positions can be optimized via gradient descent [156], their discrete combinatorial connectivity cannot be straightforwardly differentiated, preventing topology changes from emerging naturally during optimization.

Implicit representations take a fundamentally different approach. Rather than specifying the surface directly, they define a field over a spatial domain, within which the target geometry is characterized through conditions on that field, such as a level set. Different choices of the field function and the geometry condition yield representations with distinct trade-offs in expressiveness, regularity, and compatibility with downstream algorithms. Three principal scalar fields, the occupancy field (OF), the signed distance field (SDF), and the unsigned distance field (UDF), form the most widely adopted choices. Beyond them, varying the query domain, value domain, or geometry condition gives rise to vector, ray, segment, and medial representations.

Implicit fields share properties that make them naturally complementary to explicit meshes (Fig. 1). They admit gradient-based construction from diverse inputs while freely accommodating topological changes without remeshing. The resulting field can then be converted to explicit meshes via iso-surface extraction (Sec. 3), bridging optimization-friendly implicit representations to the explicit meshes widely adopted in downstream geometry tasks.

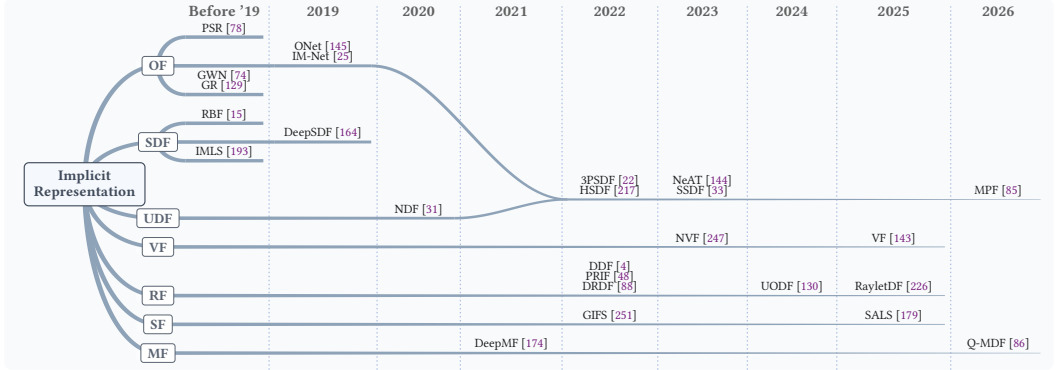


Fig. 2. Evolution of implicit representations.

Centering on this complementarity, this survey proceeds as follows. We first detail each field type under a unified framework (Sec. 2), then review iso-surface extraction algorithms (Sec. 3), followed by two downstream applications based on implicit representations: surface reconstruction from various kinds of inputs (Sec. 4) and 3D geometry generation (Sec. 5). We further benchmark point cloud surface reconstruction quantitatively, the longest-standing application of implicit fields (Sec. 6). Finally, we highlight open challenges and promising research directions that point beyond the current landscape of implicit representations and surface extraction. Note that two existing surveys are closely related to ours. Huang et al. [70] also review and benchmark point cloud surface reconstruction, but from a task-oriented perspective. Xie et al. [234] organize neural fields around network-level design choices such as conditioning, architecture, and forward mapping, adopting a network-oriented perspective. Ours takes a representation-centric view, extending beyond point cloud reconstruction to also cover image-based reconstruction and 3D geometry generation, and our benchmark additionally evaluates non-watertight geometry.

Throughout this survey, f denotes the field function in an implicit representation, including the occupancy function $f_O : \mathbb{R}^3 \rightarrow \{0, 1\}$, the signed distance function $f_S : \mathbb{R}^3 \rightarrow \mathbb{R}$, and the unsigned distance function $f_U : \mathbb{R}^3 \rightarrow \mathbb{R}_{\geq 0}$; ∇f denotes its spatial gradient, while S and M denote the underlying surface and medial axis, respectively.

2 IMPLICIT REPRESENTATIONS

We formalize an implicit representation as a pair

$$\mathcal{F} = (f: Q \rightarrow \mathcal{V}, C),$$

where f is a field function mapping a query domain Q to a value domain $\mathcal{V}$, and C defines how the target geometry $\mathcal{G} = C(f)$ is implicitly characterized by f , such as a level set of f . Different choices of f and C lead to representations with distinct trade-offs in expressiveness, regularity, and compatibility with downstream algorithms. Table 1 provides a systematic comparison, and Fig. 2 traces their chronological development. Following common usage, we use “implicit field” to refer to the representation $\mathcal{F}$ when no ambiguity arises; f denotes the underlying field function.

The field function f can be realized in two broad forms. *Continuous realizations* define a parametric function evaluable at arbitrary query points, encompassing classical basis-function models such as radial basis functions (RBFs) [71, 15, 214] and moving least squares (MLS) [82, 193, 1], neural networks [208, 198, 47, 187, 120, 244], and neural extensions that couple discrete spatial structures with neural decoders for efficiency [207, 252, 151]. They can be compact and return exact values anywhere, but each evaluation requires a forward pass through the parametric function. *Discrete*

Table 1. Comparison of Implicit Geometry Representations.

	Occupancy Field (OF)	Signed Distance Field (SDF)	Unsigned Distance Field (UDF)	Vector Field (VF/NVF)	Ray Field (RF)	Segment Field (SF)	Medial Field (MF)
Definition	Binary inside/outside indicator	Signed Euclidean distance to surface	Unsigned Euclidean distance to surface	Vector toward nearest surface point	Distance along ray to first intersection	Segment-surface intersection indicator	Inscribed ball radius at nearest medial axis point
Field Formulation	$f_0: \mathbb{R}^3 \rightarrow \{0, 1\}$	$f_S: \mathbb{R}^3 \rightarrow \mathbb{R}$	$f_U: \mathbb{R}^3 \rightarrow \mathbb{R}_{\geq 0}$	$f_V: \mathbb{R}^3 \rightarrow \mathbb{S}^2$ (VF) $f_N: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ (NVF)	$f_R: \mathbb{R}^3 \times \mathcal{D} \rightarrow \mathbb{R}_{\geq 0}$	$f_{Seg}: \mathbb{R}^3 \times \mathbb{R}^3 \rightarrow \{0, 1\}$ (+ $k_{Seg}: \mathbb{R}^3 \times \mathbb{R}^3 \rightarrow [0, 1]$)	$f_M: \mathbb{R}^3 \setminus S \rightarrow \mathbb{R}_{>0}$
Geometry Condition	$\partial(f_0 = 1)$	$\{f_S = 0\}$	$\{f_U = 0\}$	$\{D_S(f_V) = -1\}$ $\{f_N(\mathbf{x}) = 0\}$	$\{\mathbf{o} + f_R \cdot \mathbf{d} \mid f_R < +\infty\}$	$\{a + k(b - a) \mid f_{Seg} = 1\}$	$\{f_M = f_S \}$ (gives M, not S)
Representation Capability	Closed, 2-manifolds only Discont. on S (relaxed: cont.)	Closed, 2-manifolds only	Arbitrary topology	Arbitrary topology	Arbitrary topology	Arbitrary topology	2-manifolds only Closed, Discont. across S
Continuity	Non-diff. on S	Non-diff. on Γ & M	Non-diff. on S & M	f_V discont. on S & M; f_N discont. on M	Discont. where first-hit surface switches or vanishes	f_{Seg} discont. on $\partial(f_{Seg} = 1)$; (k cont. where $f_{Seg} = 1$)	Undef. on S; discont. across S
Differentiability	Non-diff. on S	Non-diff. on Γ & M	Non-diff. on S & M	Non-diff. exactly where discont.	Non-diff. exactly where discont.	Non-diff. exactly where discont.	Undef. on S; non-diff. on M
Field Properties	No metric information	$\nabla f_S = \text{outward } \mathbf{n}$; $\ \nabla f_S\ = 1$ (off Γ , M); 1-Lipschitz	$\nabla f_U \parallel \mathbf{n}$; $\ \nabla f_U\ = 1$ (off S, M); 1-Lipschitz	$f_V \parallel \mathbf{n}$; $f_V = -\nabla f_U$; $f_N \parallel \mathbf{n}$; $\ f_N\ = f_U$	$f_R = \min_d f_R$; $\nabla_a f_R \cdot \mathbf{d} = -1$	$(\nabla_a k \parallel \mathbf{n}; \nabla_b k \parallel \mathbf{n})$	$f_M \geq f_S $, on M; $\nabla f_M \perp \nabla f_S$
Geometric Queries	Inside/outside	Signed distance; sphere tracing; surf. proj. with normal	Unsigned distance; surf. proj. with normal	VF: normal direction; NVF: O(1) surf. proj.	First-hit depth and point	Edge intersection test & point	Medial sphere tracing; O(1) medial axis proj.; local thickness
Edge-Surface Test	$(f_0 - \tau)$ sign change	f_S sign change	∇f_U reversal	f_V/f_N direction reversal	Ray hit within edge	Direct query f_{Seg}	$\nabla(f_M - f_S)$ reversal

realizations instead store precomputed values at finite sample locations such as voxels, grids, or adaptive discretizations [142, 35, 49], reducing each query to a cell lookup and interpolation for much faster evaluation, but with precision bounded by the sampling resolution. Adaptive structures such as octrees [211] and sparse grids [216] alleviate memory cost yet do not remove this bound. The two forms are not mutually exclusive: a continuous field can always be sampled onto a discrete grid to produce its discretized counterpart.

Regardless of its realization, f can be obtained from existing geometry or inferred from raw observations. For existing geometry, meshes support direct field evaluation, while skeleton-based methods derive fields from skeletal primitives via distance surfaces [181] or convolution surfaces [10]. Inference from observations relies on complementary mechanisms for constraining f . *Field formulations* prescribe f as the solution of a mathematical or physics-inspired model that builds structural properties into the field definition itself, spanning interpolation and approximation energies [15, 82], numerical Poisson solvers [78, 79, 168, 61], potential-theoretic formulations including charge-distribution potentials [166] and double-layer potentials [74, 6, 105, 239], and phase-field models [106]. *Data fidelity* anchors f to measurements such as point samples, oriented normals, or images via differentiable rendering. *Field regularization* imposes differential constraints on f such as the Eikonal condition, divergence penalties, and higher-order structure [52, 8, 242]. *Shape priors* inject geometric knowledge beyond the current input, from hand-designed geometric initialization [8] to data-driven priors learned from training data [164, 145].

The remainder of this section examines specific field types. Secs. 2.1, 2.2, and 2.3 cover the three principal scalar fields, namely the occupancy field, the signed distance field, and the unsigned distance field, respectively, and Sec. 2.4 broadens the scope to vector, ray, segment and medial variants.

2.1 Occupancy Fields

The occupancy field (OF) defines a binary indicator function over $\mathbb{R}^3$, assigning 1 to points inside a shape and 0 otherwise; the surface is given by the boundary of the occupied region:

$$f_0: \mathbb{R}^3 \rightarrow \{0, 1\}, \quad C: S = \partial\{\mathbf{x} \in \mathbb{R}^3 \mid f_0(\mathbf{x}) = 1\}.$$

Due to its discontinuity, it is typically relaxed to a continuous occupancy probability $p_\theta(\mathbf{x}) \in [0, 1]$ to enable gradient-based optimization [145, 25], with the surface extracted as a level set at threshold τ . The underlying indicator can also be formulated analytically [78, 74, 129].

Notably, OF inherently assumes *closed, 2-manifold* surfaces, as the inside-outside labeling is ill-defined for open boundaries or non-manifold edges (e.g. three or more faces sharing an edge).

Moreover, even in its relaxed form, it lacks metric distance information: its values saturate away from the surface, leading to vanishing gradients and limiting geometry-aware operations such as sphere tracing and nearest-point queries.

2.2 Signed Distance Fields

The signed distance field (SDF) is a scalar field over $\mathbb{R}^3$ whose value at each point equals the signed Euclidean distance to the nearest surface point, with negative values inside and positive values outside the shape:

$$f_s: \mathbb{R}^3 \rightarrow \mathbb{R}, \quad C: S = \{ \mathbf{x} \in \mathbb{R}^3 \mid f_s(\mathbf{x}) = 0 \}.$$

Similar to OF, SDF requires a *closed, 2-manifold* surface to define the sign, but further encodes metric distance information, giving rise to useful field properties and geometric queries.

Field properties. Wherever differentiable, f_s satisfies the Eikonal equation $\|\nabla f_s\| = 1$, with the gradient equal to the outward unit normal $\mathbf{n}$ at the closest surface point. As illustrated in Fig. 3 (top left), this yields a gradient field (blue) that is continuous across the surface, with level sets (green) forming uniformly spaced offset surfaces. The field is non-differentiable at two types of loci: the medial axis M (orange dashed), where the closest surface point is non-unique and hence the gradient is undefined; and the sharp feature set $\Gamma \subset S$ (orange dot), such as edges and corners, where the surface normal is undefined. The Eikonal property implies that f_s is globally 1-Lipschitz, i.e., $|f_s(\mathbf{x}) - f_s(\mathbf{y})| \leq \|\mathbf{x} - \mathbf{y}\|$. Enforcing $\|\nabla f_s\| \approx 1$ via Eikonal regularization has become standard practice in neural SDF optimization [52].

Geometric queries. The 1-Lipschitz property guarantees that $|f_s(\mathbf{x})|$ is a conservative lower bound on the distance to S , enabling sphere tracing [56]: as shown in Fig. 3 (top right), a ray starting from $\mathbf{x}(t_0)$ along direction $\mathbf{v}$ is safely advanced by $\mathbf{x}(t_i) = \mathbf{x}(t_{i-1}) + f(\mathbf{x}(t_{i-1}))\mathbf{v}$ without overshooting. Once f_s changes sign between consecutive steps, the surface crossing is reliably bracketed and the intersection can be refined by interpolation. This technique is widely adopted in differentiable rendering pipelines [76, 113]; architectures that enforce Lipschitz continuity by design, such as 1-LNDF [34], further guarantee robustness. Additionally, the closest surface point can be recovered via projection $\Pi(\mathbf{x}) = \mathbf{x} - f_s(\mathbf{x}) \nabla f_s(\mathbf{x}) / \|\nabla f_s(\mathbf{x})\|$ (Fig. 3, bottom right), which is widely used to formulate geometric losses for fitting SDFs to raw point clouds [134].

A practical variant, the truncated SDF (TSDF), clamps values to a narrow band $f_T(\mathbf{x}) = \text{clip}(f_s(\mathbf{x}), -\delta, \delta)$, sacrificing global distance outside the truncation region but concentrating representation capacity near the surface and improving robustness to noise. It is widely adopted in real-time volumetric reconstruction and SLAM [35, 154] as well as in 3D geometry compression [210, 209, 178].

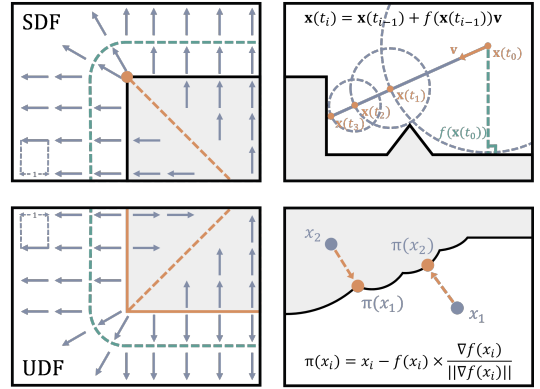


Fig. 3. Properties and applications of distance fields. Left: field properties of SDFs and UDFs: solid curves denote surface boundaries, blue arrows indicate unit gradients, green dashed curves denote offset surfaces, and orange marks non-differentiable loci, including sharp features, medial axes shown as dashed curves, and UDF surfaces shown as solid curves. Right: geometric queries, including sphere tracing and closest-point projection.

2.3 Unsigned Distance Fields

The unsigned distance field (UDF) is a scalar field over $\mathbb{R}^3$ whose value at each point equals the unsigned Euclidean distance to the nearest surface point:

$$f_U: \mathbb{R}^3 \rightarrow \mathbb{R}_{\geq 0}, \quad C: S = \{ \mathbf{x} \in \mathbb{R}^3 \mid f_U(\mathbf{x}) = 0 \}.$$

Unlike OF and SDF, UDF requires no global inside/outside classification, enabling it to represent surfaces of arbitrary topology, including open and non-manifold geometry.

Field properties. UDF shares several properties with SDF: it satisfies the Eikonal condition $\|\nabla f_U\| = 1$ wherever differentiable and is globally 1-Lipschitz, with the gradient parallel to the surface normal at the nearest point. Fig. 3 reveals the key structural difference between UDF and SDF: the UDF gradient (blue) points away from S on both sides and reverses discontinuously across the surface. Thus, UDF is non-differentiable on the surface, in addition to the medial axis (orange).

Geometric queries. The closest surface point can also be recovered via projection $\Pi(\mathbf{x}) = \mathbf{x} - f_U(\mathbf{x}) \nabla f_U(\mathbf{x}) / \|\nabla f_U(\mathbf{x})\|$ (Fig. 3, bottom right). Sphere tracing, however, is less reliable: since UDF is strictly non-negative, the sign-change criterion used by SDF is unavailable, and one must instead halt when $f_U < \tau$ for a small threshold τ . A marching step may therefore overshoot a narrow region where f_U dips below τ only briefly, missing the surface entirely.

Challenges. The non-differentiability at the zero-level set poses difficulties for both learning and surface extraction. On the learning side, neural approximations inevitably smooth the sharp minimum at S , corrupting distance values and gradients near the surface [258, 44] and producing fragmented reconstructions. On the extraction side, standard iso-surface algorithms such as Marching Cubes [128] rely on sign changes between adjacent grid vertices; since UDF is strictly non-negative, such detection fails, necessitating specialized algorithms that identify the zero-level set without sign flips.

2.4 Other Implicit Representations

For a watertight and manifold shape, the three scalar fields above are closely related (Fig. 4): OF can be obtained by thresholding SDF, $f_O(\mathbf{x}) = 1[f_S(\mathbf{x}) < 0]$, discarding all metric distance information; UDF is the absolute value of SDF, $f_U(\mathbf{x}) = |f_S(\mathbf{x})|$, removing the sign at the cost of non-differentiability at the zero-level set. Beyond these scalar fields, representations have been proposed that enrich the value domain to vectors, change the query domain from points to rays or segments, or target skeletal geometry rather than the surface. We survey each family in turn below.

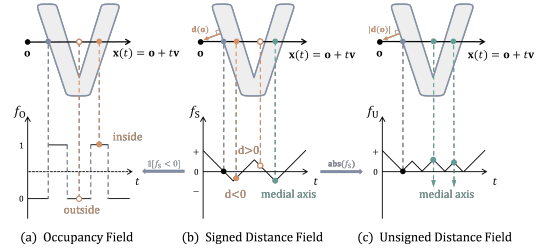


Fig. 4. Visual illustrations of OF, SDF, and UDF in 2D.

2.4.1 Vector fields. Vector fields (Fig. 5(a)) replace scalar outputs with 3D vectors pointing toward the nearest surface point, inheriting UDF's ability to represent arbitrary topology while additionally offering direct normal and projection queries.

VF [143] predicts only the direction toward the closest surface point:

$$f_V: \mathbb{R}^3 \rightarrow \mathbb{S}^2, \quad C: S = \{ \mathbf{x} \in \mathbb{R}^3 \mid D_\Phi(f_V)(\mathbf{x}) = -1 \},$$

where D_Φ denotes the flux-density operator. Since $f_V = -\nabla f_U$ away from S and M , it directly encodes the surface normal. For meshing, voxels whose discrete flux density falls below a threshold are flagged as containing S ; the eight predicted directions at each flagged voxel's vertices are then

split into two clusters by cosine similarity to assign inside–outside labels for standard iso-surface extraction.

NVF [247] predicts the full displacement toward the closest surface point:

$$f_N: \mathbb{R}^3 \rightarrow \mathbb{R}^3, \quad C: \mathcal{S} = \{ \mathbf{x} \in \mathbb{R}^3 \mid f_N(\mathbf{x}) = \mathbf{0} \},$$

whose norm equals f_U away from $\mathcal{M}$ and whose output yields $O(1)$ surface projection via $\mathbf{x} + f_N(\mathbf{x})$ in a single, differentiation-free forward pass.

2.4.2 Ray fields. Ray fields (Fig. 5(b)) lift the input from a single point to an oriented ray $(\mathbf{o}, \mathbf{d})$. In its canonical form, a ray field encodes the distance along the ray to its first surface intersection:

$$f_R: \mathbb{R}^3 \times \mathcal{D} \rightarrow \mathbb{R}_{\geq 0}, \quad C: \mathcal{S} = \{ \mathbf{o} + f_R(\mathbf{o}, \mathbf{d}) \mathbf{d} \mid f_R(\mathbf{o}, \mathbf{d}) < +\infty \}.$$

This recovers a surface point with a single query per ray. Since $f_U(\mathbf{o}) = \min_{\mathbf{d}} f_R(\mathbf{o}, \mathbf{d})$, ray fields generalize UDF while providing richer directional information; the spatial gradient further satisfies a directed eikonal property $\nabla_{\mathbf{o}} f_R \cdot \mathbf{d} = -1$ [4]. However, f_R is inherently discontinuous: small perturbations in position or direction can switch the first-hit surface, and its directional dependence precludes standard iso-surface extraction.

DDF [4] and RayDF [124] retain the full $\mathcal{D} = \mathbb{S}^2$ domain: the former predicts multiple candidate depths per ray with learned blending weights to smooth over occlusion boundaries, while the latter constrains different rays hitting the same surface point to agree in 3D. Operating on camera rays, DRDF [88] redefines the output as signed along the ray so that the field transitions continuously through zero at the surface, enabling robust extraction under prediction uncertainty, while PRIF [48] reparameterizes each ray by its perpendicular foot to the origin and directly outputs the 3D hit point, ensuring a unique input for each geometric ray. UODF [130] restricts distance queries to three orthogonal directions, enabling direct surface-point estimation along each axis and reducing interpolation errors from scalar distance values. RayletDF [226] further localizes the representation by replacing global rays with short segments originating near the surface and blending predictions via learned confidence for cross-scene generalization.

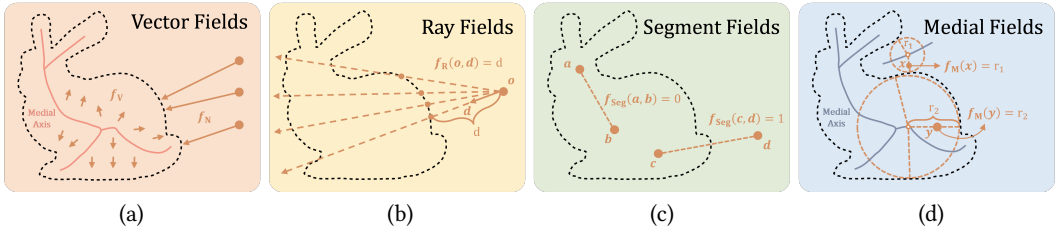


Fig. 5. Visual illustration of other implicit fields in 2D.

2.4.3 Segment fields. Segment fields (Fig. 5(c)) take a line segment rather than a single point or ray as input, querying whether it intersects the surface:

$$f_{\text{Seg}}: \mathbb{R}^3 \times \mathbb{R}^3 \rightarrow \{0, 1\}, \quad C: \mathcal{S} = \{ \mathbf{a} + k(\mathbf{a}, \mathbf{b})(\mathbf{b} - \mathbf{a}) \mid f_{\text{Seg}}(\mathbf{a}, \mathbf{b}) = 1 \},$$

where $k \in [0, 1]$ is the intersection parameter. They handle arbitrary topology and detect edge–surface intersections by a direct query of f_{Seg} , requiring no sign change or gradient reversal detection.

GIFS [251] infers the intersection parameter k from a jointly learned UDF, whereas SALS [179] directly predicts both the indicator and k in a single forward pass, further showing that $\nabla_{\mathbf{a}} k$ and $\nabla_{\mathbf{b}} k$ are parallel to the surface normal, enabling normal recovery without additional queries.

2.4.4 Medial fields. The medial field (Fig. 5(d)) targets skeletal geometry rather than the surface itself, building on the medial axis transform [11, 206]:

$$f_M: \mathbb{R}^3 \setminus S \rightarrow \mathbb{R}_{>0}, \quad C: M = \{ \mathbf{x} \in \mathbb{R}^3 \setminus S \mid f_M(\mathbf{x}) = |f_S(\mathbf{x})| \},$$

where $f_M(\mathbf{x})$ equals the length of the medial spoke through $\mathbf{x}$, connecting its nearest surface point to the medial axis. It is undefined on S , equals $|f_S|$ exactly on M , and is larger elsewhere.

Learned jointly with SDF [174], the medial field encodes local thickness and enables $O(1)$ projection onto M as well as medial sphere tracing, which improves convergence over standard sphere tracing. Since f_M is derived from SDF, it is also restricted to *closed, 2-manifold* shapes.

3 IMPLICIT FIELDS TO EXPLICIT MESHES

Given an implicit field, surface extraction produces an explicit mesh approximating the underlying geometry, bridging implicit fields to deployable meshes for rendering, simulation, and geometry processing. We focus on OF, SDF, and UDF, which have motivated the richest body of extraction algorithms; other field types either reuse similar machinery or recover surface points directly from their field queries before applying downstream meshing (Sec. 2.4).

We organize methods by field type: OF and SDF admit classical isosurfacing, while UDF requires specialized algorithms due to the absence of sign information. For each, we first describe general extraction algorithms, then their differentiable and learning-based variants.

3.1 Extraction from OF and SDF

OF and SDF both induce an inside/outside partition of space, and the surface is defined as the boundary between the two regions: the zero level set for SDF and an occupancy iso-surface for OF. SDF additionally encodes metric distance, enabling methods that exploit this richer information.

3.1.1 General Extraction Algorithms. We group methods into sign-based and metric-based, both operating on a sampled grid, and efficient strategies tailored specifically to neural fields.

Sign-based methods sample a scalar field on a grid and identify cells whose vertices exhibit mixed signs or occupancy values, indicating that the surface passes through them. They differ in how mesh vertices are placed and connected within or around these cells.

Marching Cubes (MC) [128] is the most widely used iso-surfacing algorithm. It classifies each vertex of a regular grid as inside or outside, and a precomputed lookup table maps the 2^8 possible sign configurations per cell to triangle patches, with vertex positions linearly interpolated from the field values along sign-changing edges.

MC is efficient and embarrassingly parallel, but (i) vertices are confined to grid edges, limiting fidelity for sharp features (Fig. 6, MC); and (ii) topological ambiguities arise when diagonally opposite face vertices share the same sign (Fig. 6, MC (Ambiguous)), with analogous ambiguities inside the cell body. The Asymptotic Decider [158] resolves face ambiguities via the bilinear interpolant at the face saddle point; MC33 [29] extends this to interior ambiguities; and subsequent variants [127, 93] address numerical degeneracies. Marching Tetrahedra (MT) [38] sidesteps ambiguities by subdividing each cell into tetrahedra, for which linear interpolation yields a unique intersection pattern, at the cost of more triangles and directional bias from the chosen subdivision.

Dual Contouring (DC) [77] places one vertex per sign-changing cell by minimizing a Quadric Error Function (QEF) over Hermite data (edge—surface intersections with normals). For each sign-changing edge, the four adjacent dual vertices are connected into a non-planar quadrilateral, which can be split diagonally into triangles. Because the vertex can lie anywhere inside the cell, DC preserves sharp features better than MC, but the unconstrained minimizer can drift outside the

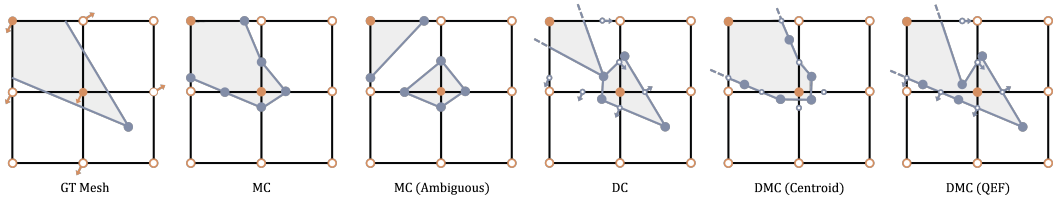


Fig. 6. Comparison of surface extraction methods in 2D. At grid vertices, filled and hollow orange dots indicate inside and outside samples, respectively, with arrows showing ground-truth normals. Hollow blue dots denote predicted edge—surface intersections; filled blue dots are the final mesh vertices. Gray shading depicts the reconstructed surface.

cell and cause self-intersections, and the one-vertex-per-cell assumption produces non-manifold output when multiple sheets traverse a single cell (Fig. 6, DC).

Dual Marching Cubes (DMC) [157, 188] mitigates both MC’s edge-constrained vertex placement and DC’s one-vertex-per-cell limitation by constructing a disambiguated MC patch graph that separates distinct sheets within each cell, placing one dual vertex per patch and stitching patches into a guaranteed manifold mesh. [157] positions each vertex at the patch centroid for robustness (Fig. 6, DMC (Centroid)), while [188] minimizes a QEF per patch to better preserve sharp features (Fig. 6, DMC (QEF)).

Metric-based methods exploit the distance value at each grid sample, which defines a ball tangent to the surface at the nearest point, with its interior guaranteed surface-free by the 1-Lipschitz property. Reach for the Spheres [191] deforms an initial mesh by minimizing a tangency-violation energy, but its topology is fixed during optimization. Reach for the Arcs [190] instead extracts tangent points on each sphere’s exposed arc into an oriented point cloud, from which a mesh of arbitrary topology is obtained via Poisson Surface Reconstruction [78].

Efficient extraction for neural fields. Evaluating a neural field on a dense grid incurs many redundant queries far from the surface. MISE [145] hierarchically refines only cells with sign changes via an octree, and McGrids [176] constructs an adaptive tetrahedral grid via Monte Carlo sampling near the surface, both substantially reducing network evaluations. For ReLU MLPs, Analytic Marching [91] and Marching Neurons [202] avoid gridding by exploiting piecewise linearity: the zero set is planar within each linear region. The former marches among neighboring regions from a seed point but may miss disconnected components; the latter traverses the network layer by layer, splitting cells along neuron hyperplanes, naturally handling arbitrary topology.

3.1.2 Differentiable and Learning-Based Variants. Two strategies integrate mesh extraction into gradient-based pipelines: making an existing algorithm differentiable, or training a network to learn the extraction directly.

Differentiable extraction layers. MeshSDF [175] keeps standard MC in the forward pass and derives analytic gradients for extracted surface samples by modeling how SDF perturbations locally move the zero crossing. Recent differentiable variants operate on discrete fields and make mesh extraction optimization-friendly by parameterizing vertex placement within local cells. DMTet [194] applies this idea to MT on deformable tetrahedral grids, TetWeave [9] to Delaunay tetrahedralizations with directional signed distances, and FlexiCubes [195] to DMC by augmenting SDF values with auxiliary extraction parameters.

Learned extraction modules. Deep Marching Cubes [101] replaces discrete sign classification with continuous occupancy probabilities, making topology selection differentiable. Neural Marching Cubes [27] redesigns tessellation templates with additional intra-cell vertices and trains a 3D CNN to predict per-cell topology and vertex positions from local context, recovering sharp edges beyond

classical MC's reach. Neural Dual Contouring [26] and Self-Supervised DC [205] replace DC's QEF solve with learned vertex prediction, the latter removing the need for ground-truth mesh supervision. A separate line replaces the fixed grid with Voronoi/Delaunay structures: VoroMesh [141] directly uses Voronoi bisector faces between oppositely labeled points as surface elements, while PoNQ [140] computes a quadric-optimal vertex per cell and connects them via Delaunay tetrahedralization with graph-cut labeling, achieving better sharp-feature preservation with fewer faces. Both are trained end-to-end on input SDF grids.

3.2 Extraction from UDF

UDF naturally supports open and non-manifold geometry that OF and SDF cannot represent. However, extraction is harder: the field is everywhere non-negative, lacking sign changes, and learned UDFs often fail to attain exact zero at the surface since smooth networks round off the non-differentiable minimum. Most methods therefore rely on UDF gradients as an additional cue.

3.2.1 General Extraction Algorithms. We group methods into three families: sign-inference methods, double-covering methods, and dual-contouring-style methods.

Sign-inference methods assign pseudo-signs to grid vertices using UDF gradients, then apply MC or its multi-label variant for extraction. MeshUDF [55] and CAP-UDF [265] select an anchor vertex per cell and compute gradient inner products with the remaining vertices to assign binary pseudo-signs; MeshUDF additionally enforces global consistency via breadth-first propagation. GeoUDF [177] instead evaluates gradient constraints for every vertex pair within each cell to jointly determine the sign configuration, selecting the best-matching MC template via a cost function. MIND [24] handles non-manifold surfaces by first constructing local two-signed fields around manifold patches, then extending them into a global multi-labeled field for multi-labeled MC extraction. All four methods rely on UDF gradients, making sign assignment unreliable near $\mathbf{M}$ where neighboring vertices may have closest points on different surface sheets.

Double-covering methods extract a double-layered shell at a small positive offset via standard MC, project it onto the true surface via a covering map, and apply min-cut to remove one layer. The min-cut assumes an orientable manifold surface; non-manifold structures or non-orientable manifolds, such as a Möbius strip, cannot be faithfully recovered. DCUDF [62] introduces this pipeline; DCUDF2 [23] adds an accuracy-aware loss and topology editing to reduce offset sensitivity.

Dual-contouring-style methods partition space into local regions, optimize one representative surface vertex per region from local tangent constraints, and recover connectivity from region dual adjacency. DualMesh-UDF [258] instantiates this on an adaptive octree, solving a QEF in each non-empty leaf from tangent planes estimated from filtered off-surface samples. VoroUDF [218] replaces the octree with an optimized Voronoi partition, updates seed positions using local tangent constraints within each Voronoi cell and derives connectivity from approximate geodesic Voronoi adjacency, thereby better preserving open boundaries, sharp features, and non-manifold geometry.

3.2.2 Learning-Based Variants. NSD [200] trains a lightweight MLP to predict per-cell sign configurations from UDF values and gradients, converting the UDF locally into a pseudo-SDF meshable by standard MC; its extension [199] adds iterative passes propagating neighborhood context to correct ambiguous cells. UNDC [26] adapts its learned DC framework to UDF, predicting dual vertices and edge occupancy from UDF values alone.

4 SURFACE RECONSTRUCTION

Recovering continuous surfaces from sensor observations such as images or point clouds is essential for downstream rendering, simulation, and editing. As discussed in earlier sections, implicit fields

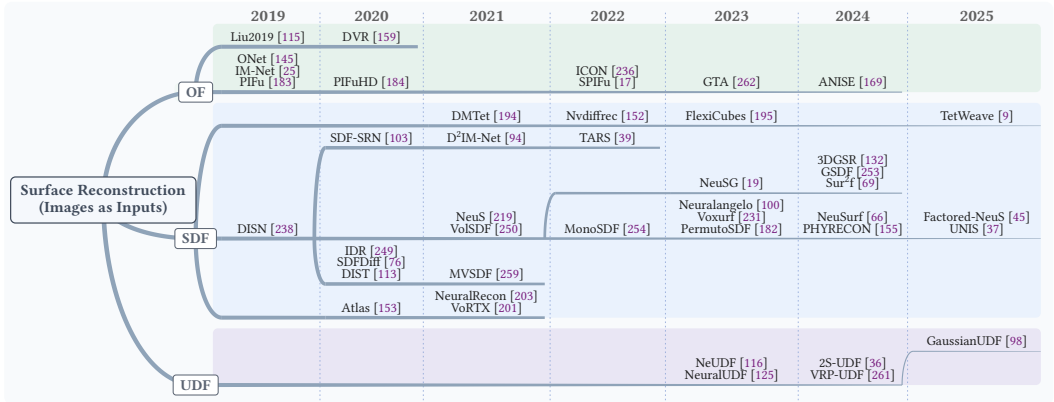


Fig. 7. Evolution of implicit surface reconstructions (2D images as inputs).

serve as a natural bridge in this pipeline: they can be reconstructed from diverse observation types and then converted to explicit meshes via iso-surface extraction.

We focus on deterministic methods that seek the single field best explaining the input; generative approaches that model a distribution over plausible shapes are covered in Sec. 5. The rest of this section is organized by input modality: **2D images** and **3D point clouds**.

4.1 2D Images as Inputs

Reconstructing a surface from 2D images is attractive because images are easy to capture, yet the geometric constraints they provide are indirect, coupling geometry with illumination, texture, and specular effects. Implicit fields are well suited to this setting because they can be differentially rendered, enabling direct supervision from image observations.

We discuss two task settings: *single-view reconstruction*, where a single photograph leaves the geometry severely under-constrained, and *multi-view reconstruction*, where cross-view consistency substantially constrains it. Fig. 7 summarizes the evolution of these methods.

4.1.1 Single-view Reconstruction. The methods reviewed below learn an encoder–decoder mapping from a single image to a neural implicit field: an encoder extracts image features that condition a coordinate-based decoder defining the field. They differ primarily in supervision: some train with paired *3D ground truth*, while others rely solely on *2D observations*. In both cases, the learned prior enables plausible completion of occluded regions for categories seen during training.

Learning with 3D supervision. When paired 3D ground truth is available, the network is trained by directly minimizing the discrepancy between predicted and reference field values. The central design choice is then what image features the encoder extracts to condition the field. The earliest methods, such as ONet [145] and IM-Net [25], encode the image into a single global latent code, but this tends to oversmooth local detail. To recover finer geometry, PIFu [183] introduces pixel-aligned conditioning for clothed human reconstruction, projecting each 3D query into the image and sampling the corresponding local feature, with subsequent work enriching conditioning or injecting domain-specific priors [184, 262, 236, 17]. For general objects, DISN [238] combines global and local features and learns the 2D projection of each 3D query, removing the need for known camera parameters. Other strategies include coarse-to-fine refinement [94] and part-level assembly [169].

Learning with 2D supervision. When paired 3D supervision is unavailable, the network is instead trained through differentiable image formation: points sampled along camera rays are

queried and the resulting predictions are compared against observed silhouettes or colors. Liu et al. [115] instantiate this idea for OFs under multi-view silhouette supervision by aggregating occupancy along each ray into a foreground probability. DVR [159] finds the first ray-surface intersection of the OF and evaluates a texture field there, deriving an analytic depth gradient via implicit differentiation so that both mask and RGB observations supervise geometry. SDF-SRN [103] switches to an SDF with SRN-style ray marching and replaces binary masks with silhouette distance transforms, making every pixel contribute geometric signal. TARS [39] extends this SDF-based rendering with a topology-aware deformation into a canonical space, enabling category-level correspondence learning under 2D supervision alone.

4.1.2 Multi-view Reconstruction. With multiple views available, the observed geometry is often well determined, and the focus shifts to how best to recover it. Most methods assume known camera poses, although some jointly refine or even estimate them during reconstruction. We group multi-view approaches into three families: *feed-forward prediction* from the input views, *per-instance optimization* that fits the field directly to the observed images, and *hybrid representations* that couple an implicit field with an explicit geometric proxy.

Feed-forward prediction. These methods fuse image features into a voxel grid to predict a TSDF in a single forward pass [153, 203, 201], differing mainly in the fusion strategy. They are fast and generalizable, but their fixed voxel resolution limits geometric fidelity.

Per-instance optimization. In contrast, per-instance methods fit an implicit field to each test instance so that its differentiable renderings explain the input views. Distance fields are usually preferred over OFs for their richer geometric signal. This per-instance fitting is slower than feed-forward prediction, but usually yields higher fidelity. The main difference then lies in how the implicit field is differentially rendered into images.

Surface rendering supervises an SDF at the first ray-surface intersection, typically located by sphere tracing; methods differ mainly in how this intersection is differentiated. IDR [249] and MVSDf [259] backpropagate through the zero-level-set hit point via implicit differentiation. SDFDiff [76] restricts the gradient to the local grid neighborhood around the hit, while DIST [113] introduces a practical differentiable sphere-tracing layer for neural SDFs with efficient forward and backward approximations.

SDF-based volume rendering brings SDF geometry into the NeRF-style volume rendering framework [147], converting SDF values along each ray into opacity weights and integrating color over all samples. Two concurrent works established this paradigm: VoSDF [250] maps the SDF to density with a Laplace CDF and derives an opacity approximation error bound to guide sampling, whereas NeuS [219] uses a logistic-based mapping and derives first-order unbiased rendering weights that peak exactly at the zero level set. UNIS [37] unifies these mappings in a broader family and characterizes the conditions for first-order unbiasedness. Follow-up work extends this paradigm, e.g., with stronger geometric priors for under-constrained scenes [254, 66], more efficient spatial parameterizations [100, 182, 231], and richer objectives such as physical plausibility or inverse rendering [155, 45].

To handle non-watertight surfaces, *UDF-based volume rendering* extends this idea to unsigned distances, with the central challenge being their conversion into reliable ray weights. NeuralUDF [125] and NeUDF [116] initiated this line with analytic weight formulations, and follow-up work revisits weight construction through staged refinement [36] or learned priors [261].

Mesh rendering instead optimizes a discrete field, extracts an explicit mesh with a differentiable iso-surfacing layer, and supervises that mesh through differentiable rasterization. Nvdiffrnc [152] instantiates this pipeline with DMTet [194] and jointly optimizes geometry, materials, and lighting. FlexiCubes [195] extends DMC on regular cube grids with flexible vertex positioning and per-cell

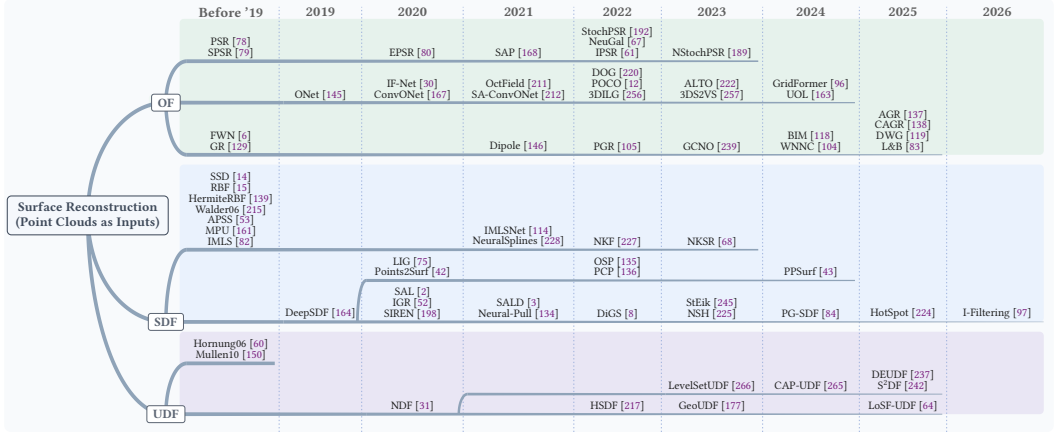


Fig. 8. Evolution of implicit surface reconstructions (3D point clouds as inputs).

topology control, whereas TetWeave [9] replaces the predefined tetrahedral grid of DMTet with on-the-fly Delaunay tetrahedralization for adaptive resolution.

Comparison of rendering strategies. Surface rendering is geometrically direct, but each ray supervises only one surface point, making optimization sensitive to self-occlusion, thin structures, and abrupt depth changes. Volume rendering distributes supervision over all samples along a ray, greatly improving optimization robustness, but requires many field evaluations and carefully designed field-to-weight mappings. Mesh rendering optimizes geometry through an extracted mesh and differentiable rasterization, making it efficient and directly amenable to joint material and lighting estimation, but its fidelity is bounded by the underlying grid resolution and iso-surfacing quality.

Hybrid representations. Hybrid methods co-optimize the implicit field with an explicit proxy that provides dense geometric cues or efficient rendering, while the implicit field supplies a smooth, extractable surface. Sur²f [69] pairs an SDF with a deformable surrogate mesh, synchronizing them through a shared neural shader and using the mesh for surface-guided volume sampling. More recently, 3D Gaussians [81] have become a popular proxy owing to their fast differentiable rasterization. Several methods couple 3D Gaussians with a neural SDF, using the Gaussians to provide geometric priors or rendering supervision and the SDF to regularize the underlying geometry [19, 132, 253]; GaussianUDF [98] extends this hybrid paradigm to open surfaces by pairing Gaussians with a UDF.

4.2 3D Point Clouds as Inputs

Point clouds are a fundamental 3D sensing modality, but their discreteness, lack of connectivity, and missing or inconsistent orientation make surface reconstruction ill-posed. Implicit fields are well suited to this setting because point samples naturally provide local constraints on the field function f , e.g., $f(\mathbf{p}) = 0$ for surface samples and $\nabla f(\mathbf{p}) = \mathbf{n}$ for oriented points.

We organize these methods by how they define and constrain f : *classical implicit field methods* define f through explicit mathematical or physics-inspired formulations, often gaining stronger built-in geometric regularity; *neural implicit field methods* parameterize f with coordinate networks, gaining learnable flexibility and data-driven shape priors while relying more on data and regularization. Fig. 8 summarizes the evolution of these methods.

4.2.1 Classical Implicit Field Reconstruction. Classical implicit methods define f through prescribed mathematical or physics-inspired structures rather than a learned coordinate network. These structures give explicit control over regularity and often yield robust, watertight reconstructions, but many formulations are well-posed only when globally consistent normal orientation is available.

Oriented-point methods. When globally consistent normals are available, reconstruction is much better conditioned: normals provide both local surface orientation and the sign cues needed by many classical formulations. The main distinction is then how the oriented input determines f : as the solution of a global variational problem, by interpolation or approximation within a prescribed function family, or through an integral formula induced by the input.

Variational PDE methods. These methods define f implicitly as the minimizer of a global variational objective. Poisson Surface Reconstruction (PSR) [78] is the canonical example: it interprets the oriented normals as samples of the gradient of an indicator function, interpolates them into a dense vector field $\mathbf{V}$, and solves the corresponding Poisson problem to recover an indicator-like field whose gradient best approximates $\mathbf{V}$. The global solve produces an inherently watertight surface, but constraining only ∇f leaves the absolute field value unanchored, requiring a post-hoc iso-value choice. Screened PSR [79] and SSD [14] both add position fidelity, the former within the original indicator-function framework and the latter by directly targeting a smooth approximate SDF. Follow-up work extends the same variational route with envelope constraints from visual or depth hulls [80], uncertainty estimation through stochastic reformulations [192, 189], or learned spatially adaptive basis functions while still solving for the coefficients through a linear system [67].

Interpolation and approximation methods. These methods prescribe an explicit functional form for f and determine it from the samples, either by global interpolation through the data or by local approximation near each query point.

Global interpolation methods construct f as a weighted sum of basis kernels centered at the samples and solve for the coefficients; the choice of kernel then governs both the smoothness of the reconstruction and the computational cost. RBF [15] places a globally supported polyharmonic kernel at each sample and fits the field by enforcing value constraints on and off the surface, yielding a smooth interpolant at $O(N^3)$ cost. HRBF [139] replaces the off-surface constraints with direct Hermite conditions on both values and gradients. Neural Splines [228] substitutes the hand-designed kernel with the arc-cosine kernel of infinitely wide shallow ReLU networks, drawing the prior from the corresponding neural RKHS. Walder et al. [215] switch to compactly supported multi-scale bases with a global regularizer, yielding sparse systems that scale to larger inputs.

Local approximation methods instead evaluate f at each query point from fits to nearby samples. IMLS [193, 82] computes a locality-weighted average of signed distances to tangent planes at nearby oriented samples. APSS [53] replaces tangent planes with algebraic spheres for better stability under high curvature. MPU [161] blends local quadratic approximants on an adaptive octree through partition-of-unity weights.

Recent learned variants preserve this explicit evaluation structure while learning components of the interpolation or approximation family. NKF [227] learns a global kernel prior and reconstructs by kernel ridge regression, while NKSR [68] replaces global kernels with compactly supported ones and adds gradient fitting, scaling to large scenes. IMLSNet [114] learns the local MLS parameterization, predicting the octree scaffold and primitives from which the field is evaluated in closed form.

Integral methods. These methods define f directly by integrating contributions from the oriented input rather than solving for unknown coefficients. GWN [74] extends the classical winding number to oriented triangle soups, yielding a continuous inside–outside field that degrades gracefully under gaps and self-intersections. FWN [6] transfers this idea to oriented point clouds via a hierarchical

double-layer potential approximation:

$$f(\mathbf{q}) = \sum_{i=1}^m \frac{a_i (\mathbf{p}_i - \mathbf{q}) \cdot \hat{\mathbf{n}}_i}{4\pi \|\mathbf{p}_i - \mathbf{q}\|^3},$$

where $\mathbf{p}_i$, $\hat{\mathbf{n}}_i$, and a_i are the position, outward unit normal, and area weight of each oriented sample. GR [129] derives a related field from a modified Gauss formula, but regularizes the near-surface singularity so that the field behaves like an indicator far from the surface and like an SDF near it.

Unoriented-point methods. Without reliable normals, the main difficulty is recovering consistent sign information. Existing methods either orient the input first or estimate orientation jointly with reconstruction; early work also explored postponing signing to a post-processing step [60, 150].

Orientation as preprocessing. The standard pipeline follows Hoppe et al. [59]: estimate unoriented normals by local PCA, then propagate a consistent orientation along an MST before applying any oriented solver. PCPNet [54] improves the local normal estimates with a learned patch encoder, while Dipole [146] replaces MST traversal with dipole-based long-range propagation. More recent work treats orientation as an inverse problem on the winding-number field, optimizing candidate normals until the induced field satisfies desired structural properties. GCNO [239], BIM [118], and WNNC [104] respectively encourage approximate binarity, low Dirichlet energy, or gradient-normal consistency of the induced field.

Joint orientation and reconstruction. These methods fold orientation into the reconstruction objective itself, so that the field and the normals are coupled and solving for one constrains the other. IPSR [61] extends Screened PSR by alternating its global solve with normal re-estimation from the current surface. PGR [105] extends GR by treating the per-point surface elements as unknowns and solving for them jointly with the indicator field in one linear system; AGR [137] and CAGR [138] augment PGR with anisotropic fundamental solutions that reduce underdeterminedness, especially on thin structures. DWG [119] alternates between recomputing a screened winding field and updating normals from its level-set gradients until both stabilize. VIPSS [71] works in the same kernel function space as the classical oriented fitting methods above but treats the normals as free variables, finding the smoothest interpolant through the input points by minimizing a global biharmonic energy that couples all normals into a consistent configuration. NN-VIPSS [232] localizes VIPSS to natural neighborhoods, scaling to much larger point sets.

4.2.2 Neural Implicit Field Reconstruction. Neural implicit field methods instead represent f as a coordinate network. We group them by how this network is obtained: *learning-based prediction*, *per-instance optimization*, and *hybrid methods*.

Learning-based prediction. These methods train an encoder together with the coordinate decoder on shape datasets, so reconstruction reduces to a single forward pass at test time. They differ mainly in how the input conditions the decoder.

Spatial feature conditioning. The simplest version conditions the decoder on a single global latent code, as in ONet [145], but the fixed-size bottleneck limits fine detail. Later work therefore conditions each query on spatially organized features. IF-Net [30] voxelizes the input, learns multi-scale 3D features, and interpolates them at query points before decoding. ConvONet [167] instead projects point features onto canonical planes or volumetric grids and refines them convolutionally, yielding a scalable framework for large scenes. Subsequent work pushes the same idea toward adaptive octree features [211, 220], point-native latents [12], token-based spatial latents for transformer-style decoding [256, 257], and explicit point-grid interaction schemes [222, 96].

A parallel line extends IF-Net’s multi-scale encoding to non-watertight geometry. NDF [31] regresses unsigned distances and extracts surfaces by gradient-based projection. To enable direct

mesh extraction on open surfaces, HSDF [217] additionally predicts a sign field, while GIFS [251] predicts a segment field with an auxiliary UDF.

Query-centric feature conditioning. Instead of sampling from a pre-built spatial structure, these methods construct features from the local neighborhood of each query point, naturally adapting to varying density. Points2Surf [42] decomposes SDF prediction into a local patch-based distance regressor and a global sign classifier; PPSurf [43] extends it by fusing global context with query-local priors through attention. For open surfaces, GeoUDF [177] aggregates unsigned distances to tangent planes induced by local quadratic patches, while LoSF-UDF [64] learns a local UDF prior from synthetically generated analytic patches. SALS [179] applies the same local-query conditioning idea to a segment field, enabling direct mesh extraction on surfaces with arbitrary topology.

Per-instance optimization. These methods fit a neural field directly to a single point cloud, without dataset-level training. The main distinction is where the fitting signal is imposed: directly on the field, or indirectly through projections onto the current surface.

Fitting via direct field supervision. Methods in this family supervise the neural field f by placing losses directly on its output values and/or their spatial derivatives at sampled points.

IGR [52] establishes the foundational paradigm: on-surface zero-value constraints anchor the field to observations, while an Eikonal regularizer $\|\nabla f\| \approx 1$ throughout the domain encourages global SDF behavior. SIREN [198] replaces the ReLU MLP with sinusoidal activations whose derivatives at any order remain non-degenerate, enabling exploitation of higher-order differential properties. Building on this route, DiGS [8], StEik [245], and NSH [225] all introduce second-order geometric regularization with different goals: DiGS penalizes the Laplacian to stabilize fitting on unoriented inputs; StEik replaces the Laplacian with a normal-direction second-order term to avoid over-smoothing; NSH enforces a near-surface singular-Hessian prior to suppress ghost geometry. HotSpot [224] augments the objective with a screened Poisson term, turning the necessary Eikonal constraint into an asymptotically sufficient one.

An orthogonal sign-agnostic line fits SDFs from unsigned data. SAL [2] regresses $|f|$ to unsigned distance values and relies on sphere initialization to resolve the sign ambiguity, while SALD [3] further incorporates near-surface derivative information of the unsigned distance function to improve sample efficiency and sharp-feature recovery.

Beyond SDFs, DEUDF [237] targets high-fidelity UDF fitting with normal-alignment cues and adaptive near-surface regularization, while S²DF [242] replaces UDF with a differentiable scaled-squared distance representation paired with a Monge–Ampère-type regularizer.

Fitting via projection-based supervision. Neural-Pull [134] supervises the field indirectly through its induced surface projection: it projects each query onto the zero level set along the predicted gradient by the predicted distance, and minimizes the gap between the projected point and its nearest neighbor in the input. This pulling loss implicitly couples field values with gradient directions without an explicit Eikonal term. IFiltering [97] extends this with an implicit bilateral filter that extends to non-zero level sets, enforcing inter-level-set geometric consistency to regularize the zero level set while preserving high-frequency details. Follow-up works adapt the pulling mechanism to UDFs [265, 266], extending this route to open and non-manifold surfaces.

Bridging the direct and projection-based routes, PG-SDF [84] combines losses from both and addresses initialization sensitivity through point-guided homotopy optimization.

Hybrid methods. Hybrid methods combine learned priors with per-instance optimization to balance generalization and instance-specific fidelity. DeepSDF [164] trains a shared SDF decoder via an auto-decoder framework with per-shape latent codes; at test time a new code is fitted to observations while staying close to the learned shape manifold. LIG [75] extends this to large scenes via overlapping local latent codes. SA-ConvONet [212] initializes from a ConvONet prediction and applies sign-agnostic test-time optimization to resolve sign conflicts. OSP [135] and PCP [136] train

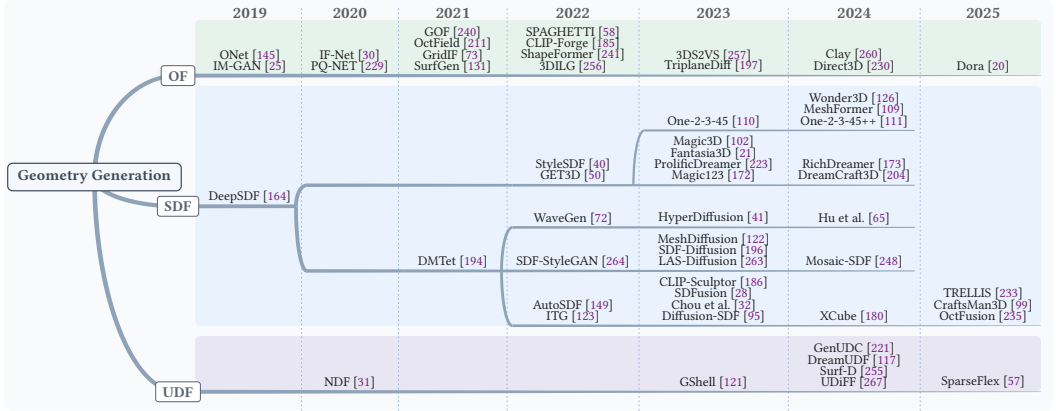


Fig. 9. Evolution of implicit geometry generation.

auxiliary prior networks and incorporate them as differentiable supervision during per-instance SDF optimization, improving robustness under severe sparsity and noise.

5 3D GEOMETRY GENERATION

The reconstruction methods of Sec. 4 seek a single implicit field that best explains the observations. This deterministic formulation works well under strong conditioning but tends to average over plausible modes as constraints weaken, producing overly smooth surfaces. Geometry generation addresses this limitation by modeling a distribution over implicit fields, enabling diverse, detail-rich samples even from weak conditioning such as a single image or a text prompt. The scarcity of large-scale 3D shape collections and the high dimensionality of implicit fields have led to two major lines of work.

2D-lifted methods bypass the data bottleneck by constraining 3D generation through supervision in 2D image space. **Native 3D methods** learn the shape distribution directly from 3D data, tackling dimensionality through compact latent representations. Fig. 9 summarizes the evolution of these methods.

5.1 2D-Lifted Generation

2D image priors support 3D generation either by directly supervising a 3D representation in image space or by first generating intermediate views that are then used for 3D reconstruction. We review three regimes: *feed-forward generation with adversarial supervision*, *per-instance optimization with diffusion priors*, and *feed-forward reconstruction from diffusion-generated views*.

5.1.1 Feed-forward Generation with Adversarial Supervision. The earliest 2D-lifted methods train a shared generator under adversarial image supervision: the generator produces an implicit field, a differentiable renderer maps it to images, and a discriminator scores the renderings. Representative works progressively shift the underlying representation from NeRF-based OFs [240] to SDFs with volume-rendered features [40], and further to DMTet-extracted meshes supervised through rasterization [50]. These models are efficient at inference, but adversarial training is prone to mode collapse and requires category-specific image collections, restricting open-domain generation.

5.1.2 Per-instance Optimization with Diffusion Priors. These methods fit a 3D representation to each input by distilling frozen diffusion priors at test time. DreamFusion [171] introduced this regime with Score Distillation Sampling (SDS) on a NeRF, enabling open-domain 3D generation

from text without any 3D training data. Follow-up work shifts the underlying representation toward surfaces: Magic3D [102] and Fantasia3D [21] adopt DMTet for explicit mesh extraction, while DreamUDF [117] extends the regime to open surfaces via UDF. Concurrent efforts improve distillation quality through variational formulations [223] or mitigate the Janus problem with 3D-aware diffusion priors [112, 172, 204, 173]. Despite broad applicability, each result still requires hundreds to thousands of optimization steps.

5.1.3 Feed-forward Reconstruction from Diffusion-Generated Views. Rather than distilling diffusion priors directly into a 3D field, this regime first synthesizes multi-view intermediates and then reconstructs 3D from them. Building on Zero-1-to-3 [112], subsequent methods refine this generate-then-reconstruct paradigm by improving multi-view synthesis, downstream 3D reconstruction, or both [110, 126, 109, 111].

5.2 Native 3D Generation

Native 3D generation learns the shape distribution directly from 3D data itself, rather than constraining 3D generation through 2D image priors or rendered image supervision. Its main challenge is scalability: the parameter space of an implicit representation is high-dimensional, and dense volumetric discretizations grow cubically with resolution. Existing methods mainly differ in where the generative model operates. *Direct generation* works in the original field space, whereas *latent generation* first compresses shapes into a lower-dimensional representation and models the distribution there.

5.2.1 Direct Generation. Direct generation applies the generative model directly to the original 3D representation without first compressing shapes into a latent code.

Adversarial methods. These methods differ mainly in where the discriminator operates. DMTet [194] extracts an explicit mesh from an SDF on a tetrahedral grid and supervises the mesh surface adversarially. SurfGen [131] projects the zero iso-surface onto the unit sphere and applies a spherical CNN discriminator. SDF-StyleGAN [264] stays entirely in the implicit domain, applying global and local discriminators to SDF values and gradients.

Diffusion and flow-matching methods. These methods replace adversarial supervision with diffusion or flow matching. One route operates directly on SDF volumes. SDF-Diffusion [196] uses a cascaded pipeline that first generates a low-resolution field and then super-resolves it patch by patch. LAS-Diffusion [263] adopts a two-stage occupancy-then-SDF pipeline with local attention for sketch-conditioned generation. Mosaic-SDF [248] departs from dense grids, representing each shape with a sparse set of local SDF patches near the boundary and modeling them with a transformer-based flow-matching network. Another route couples diffusion with structured mesh-related representations. MeshDiffusion [122] generates watertight meshes by denoising the SDF values and vertex deformations of a DMTet grid. GShell [121] and GenUDC [221] further handle open shapes through different representations: GShell defines a manifold SDF on a watertight shell enclosing the target, while GenUDC uses an Unsigned Dual Contouring representation that separates face topology from vertex geometry.

5.2.2 Latent Generation. Latent generation shifts distribution modeling from the full 3D field to a compact latent space. Shape codes are obtained either by *amortized encoding*, which learns a shared encoder-decoder over a shape collection, or by *per-shape derivation*, which obtains each code independently.

Amortized encoding. Early methods such as ONet [145] and OctField [211] adopt a single-stage VAE framework, regularizing the latent space toward a unit Gaussian for direct sampling. Yet a unimodal prior poorly captures multi-modal shape distributions, motivating two-stage pipelines

that first train an autoencoder for reconstruction and then learn the latent distribution with a more expressive model. Early two-stage methods explore increasingly structured codes, from global vectors [25] to spatially localized grids [73] and part-level assemblies [58, 229], with text-conditioned variants leveraging CLIP features [185, 123].

With diffusion replacing these earlier priors, the latent representation has continued to evolve. Early methods compress voxelized SDFs into compact spatial latent grids [95, 28] or into global latent vectors via modulation-based SDF-VAEs [32]. Subsequent work replaces the dense grid either with unstructured latent vector sets [257], a design later scaled and refined in [260, 20, 99], or with sparse spatial structures such as octrees and sparse voxel grids [235, 180, 233], with SparseFlex [57] further supporting open surfaces through sparse FlexiCubes. Other directions include triplane-factorized latents [230] and extending the underlying field to UDFs for arbitrary-topology generation [255]. A parallel discrete branch quantizes latents into codebook tokens for autoregressive generation, exploring various spatial tokenization strategies [241, 149, 256], with CLIP-Sculptor [186] extending this to zero-shot text-to-shape synthesis.

Per-shape derivation. Unlike amortized encoding, the following methods derive each shape’s latent independently. One line decomposes TSDF grids into wavelet coefficients and applies diffusion in the wavelet domain [72, 65], with UDiFF [267] extending this to UDFs for open-surface generation. Triplane Diffusion [197] fits each shape to tri-plane feature maps and trains a 2D diffusion model on those maps. HyperDiffusion [41] takes a more direct route, diffusing over the full weights of per-shape overfit MLPs.

6 BENCHMARKING IMPLICIT REPRESENTATION-BASED 3D SURFACE RECONSTRUCTION FROM 3D POINT CLOUDS

Point cloud surface reconstruction is the longest-standing application of implicit fields, studied through both classical and neural formulations as reviewed in the preceding sections. Recent benchmarking efforts [70] have advanced standardized evaluation, yet their focus on watertight geometry leaves non-watertight surfaces with open boundaries and non-manifold edges underexplored. To fill this gap, we benchmark 19 representative methods across four complementary scenarios covering both object-level geometry (watertight and non-watertight) and scene-level data (synthetic and real-scanned) with varying point cloud densities and additional noise levels for object-level shapes.

Table 2. Quantitative results on **non-watertight object** reconstruction. Only open-surface methods are evaluated. **Bold** and underline mark the best and second-best results, respectively.

Method	Sparse (3K pts)				Clean (30K pts)				Dense (100K pts)				Noise $\sigma=0.002\ell$				Noise $\sigma=0.005\ell$			
	CD↓	F1↑	NC↑	HD↓	CD↓	F1↑	NC↑	HD↓	CD↓	F1↑	NC↑	HD↓	CD↓	F1↑	NC↑	HD↓	CD↓	F1↑	NC↑	HD↓
NDF [31]	8.726	62.73	91.13	21.18	7.293	71.42	95.88	18.53	8.160	64.00	95.06	21.61	9.019	49.89	<u>94.57</u>	10.87	21.40	3.000	<u>89.80</u>	20.41
GIFS [251]	7.669	69.25	89.35	13.32	6.514	74.70	95.25	7.690	7.949	55.90	93.34	9.888	10.26	28.96	93.52	9.993	21.46	3.400	86.26	19.20
GeoUDF [177]	5.248	84.23	93.60	7.678	4.623	90.67	<u>97.26</u>	11.50	4.536	90.77	<u>97.14</u>	5.058	7.233	64.75	91.37	7.438	18.10	6.130	81.96	18.54
LevelSetUDF [266]	7.143	73.38	90.36	16.67	<u>4.090</u>	<u>90.86</u>	96.58	<u>4.344</u>	4.726	90.27	96.59	11.05	5.545	82.92	94.46	5.390	13.15	19.21	86.45	<u>13.98</u>
CAP-UDF [265]	8.504	67.51	85.07	20.09	5.116	89.46	96.52	13.87	5.386	88.89	95.84	18.00	9.478	79.11	92.52	20.99	17.31	25.87	76.63	37.16
SALS [179]	<u>6.637</u>	<u>76.34</u>	82.77	<u>11.96</u>	4.314	89.52	94.05	5.163	<u>4.108</u>	<u>90.97</u>	95.84	<u>4.407</u>	6.546	72.73	72.28	8.042	<u>11.31</u>	<u>35.36</u>	58.18	16.01
S ² DF [242]	11.30	57.17	<u>92.53</u>	25.28	4.004	91.59	98.24	4.337	3.971	91.82	98.43	4.224	<u>5.724</u>	<u>81.62</u>	95.26	<u>6.082</u>	8.478	54.21	90.69	9.779

6.1 Experimental Setup

We evaluate 19 methods spanning classical fields, learning-based neural prediction, and per-instance neural optimization across four scenarios: *watertight objects*, *non-watertight objects* containing open boundaries and non-manifold topology, *synthetic scenes* with clean ground truth, and *real-scanned scenes* with complex topologies and noise.

Methods are grouped by input configuration (oriented versus unoriented), with some evaluated under both settings; unoriented methods are further divided into closed-surface and open-surface categories, of which only the latter is applicable to non-watertight and real-scanned data.

For each object, we generate point clouds at three densities (3K, 30K, and 100K points, termed Sparse, Clean, and Dense) and two noisy variants by adding isotropic Gaussian noise with standard deviation $\sigma \in \{0.002\ell, 0.005\ell\}$ to the 30K configuration, where ℓ is the bounding-box diagonal. Scene point clouds are sampled at 500 and 1000 points per m^2 , yielding 330 evaluation instances in total. All methods are compared using Chamfer Distance (CD), F-Score (F1), Normal Consistency (NC), and Hausdorff Distance (HD). Full details are provided in the supplementary material.

6.2 Results and Analysis

Oriented normals remain a powerful but no longer indispensable prior. Under clean, dense conditions the best unoriented methods match or even surpass neural oriented methods on watertight objects (Table 4), yet classical field methods retain a unique robustness advantage under noise and at scene scale (Table 3): SPSR leads across nearly all metrics in these regimes, as its prescribed functional structure and global variational solve consider all points simultaneously, making it inherently resilient to local perturbations in a way that pointwise neural fitting cannot easily replicate.

Open-surface methods match closed-surface methods on watertight geometry while naturally extending to open topology. Within the unoriented group on watertight objects (Table 4), open-surface methods such as S^2DF and GeoUDF achieve comparable accuracy to closed-surface methods under most conditions, while additionally handling non-watertight topology where closed-surface formulations are structurally inapplicable.

Data sufficiency governs the learning-versus-optimization trade-off. Under sparse input (Tables 4 and 2), per-instance optimization methods such as S^2DF degrade sharply as too few constraints remain to resolve surface geometry, while learning-based methods such as GeoUDF maintain reasonable accuracy thanks to pretrained shape priors. The advantage reverses at higher densities, where per-instance optimization exploits fine-grained local detail more fully than fixed feed-forward predictors.

Table 3. Quantitative results on **synthetic and real-scanned scene** reconstruction. For synthetic scenes, methods are divided into oriented-input, unoriented closed-surface, and unoriented open-surface blocks; for real-scanned scenes, only open-surface methods are evaluated. Within each block, **bold** and underline mark the best and second-best results, respectively.

Method	Input	500 pts/m ²				1000 pts/m ²			
		CD↓	F1↑	NC↑	HD↓	CD↓	F1↑	NC↑	HD↓
<i>Synthetic scene</i>									
SPSR [79]	Ori	7.231	62.14	96.32	7.547	7.143	62.66	96.76	7.418
IGR [52]	Ori	8.684	<u>58.35</u>	95.81	16.33	<u>7.755</u>	<u>59.48</u>	95.94	10.21
SIREN [198]	Ori	8.118	53.16	<u>96.23</u>	<u>7.652</u>	8.308	52.36	<u>96.19</u>	8.434
IGR [52]	Unori	17.33	38.61	81.12	51.09	15.25	39.84	86.34	45.03
SAP (Opt) [168]	Unori	9.223	53.22	81.54	19.44	8.180	54.67	84.10	9.047
POCO [12]	Unori	9.190	50.58	92.03	<u>14.46</u>	9.299	53.41	91.63	18.33
DiGS [8]	Unori	8.706	54.32	<u>94.41</u>	14.92	8.705	54.19	94.64	14.39
NSH [225]	Unori	19.16	39.69	86.77	46.89	10.41	50.43	91.90	22.99
StEik [245]	Unori	7.939	56.15	95.43	8.612	<u>8.404</u>	<u>54.26</u>	<u>94.56</u>	<u>10.76</u>
PG-SDF [84]	Unori	9.607	52.18	87.97	17.94	9.505	52.67	87.93	17.21
I-Filtering [97]	Unori	9.270	<u>55.38</u>	92.29	16.29	10.61	53.38	90.75	27.02
NDF [31]	Unori	16.15	30.41	89.25	80.03	18.98	23.67	87.17	86.96
GIFS [251]	Unori	9.301	44.04	91.95	9.883	9.891	40.46	91.33	11.64
LevelSetUDF [266]	Unori	7.996	58.54	93.79	9.155	<u>7.808</u>	<u>58.86</u>	<u>93.98</u>	8.520
CAP-UDF [265]	Unori	<u>7.744</u>	<u>58.70</u>	<u>93.90</u>	8.417	7.881	58.81	<u>93.96</u>	8.676
SALS [179]	Unori	7.830	56.66	85.43	<u>8.255</u>	7.831	56.66	85.47	<u>8.271</u>
S ² DF [242]	Unori	7.283	61.45	95.67	7.813	7.290	61.36	95.77	7.837
<i>Real-scanned scene</i>									
NDF [31]	Unori	6.885	67.16	91.08	7.439	7.059	62.14	90.55	8.274
GIFS [251]	Unori	6.889	71.78	90.37	8.239	7.874	57.87	88.40	8.446
LevelSetUDF [266]	Unori	<u>3.685</u>	<u>95.74</u>	<u>92.62</u>	<u>4.084</u>	<u>3.784</u>	<u>94.79</u>	<u>92.36</u>	<u>4.626</u>
CAP-UDF [265]	Unori	3.805	95.31	92.07	4.255	3.852	94.66	92.11	4.684
SALS [179]	Unori	3.952	93.78	87.95	4.766	3.880	94.22	88.92	4.637
S ² DF [242]	Unori	3.599	96.16	93.44	3.990	3.628	95.74	93.79	4.171

Table 4. Quantitative results on **watertight object** reconstruction. The three blocks correspond to oriented-input, unoriented closed-surface, and unoriented open-surface methods. Within each block, **bold** and underline mark the best and second-best results, respectively.

Method	Input	Sparse (3K pts)				Clean (30K pts)				Dense (100K pts)				Noise $\sigma=0.002\ell$				Noise $\sigma=0.005\ell$			
		CD↓	F1↑	NC↑	HD↓	CD↓	F1↑	NC↑	HD↓	CD↓	F1↑	NC↑	HD↓	CD↓	F1↑	NC↑	HD↓	CD↓	F1↑	NC↑	HD↓
SPSR [79]	Ori	10.92	62.66	89.64	<u>24.75</u>	<u>5.149</u>	<u>86.25</u>	<u>96.46</u>	<u>6.176</u>	<u>7.335</u>	<u>87.42</u>	<u>97.76</u>	<u>19.85</u>	<u>5.554</u>	83.84	94.33	6.112	6.839	<u>71.46</u>	<u>90.19</u>	7.530
IGR [52]	Ori	6.444	79.52	94.06	11.28	4.982	87.88	97.20	6.126	4.775	88.67	98.03	4.916	11.69	<u>83.75</u>	94.64	<u>15.17</u>	14.34	72.70	90.51	<u>22.35</u>
SIREN [198]	Ori	<u>9.412</u>	<u>70.94</u>	<u>93.96</u>	31.89	9.028	78.64	96.38	36.46	9.682	78.96	96.34	27.28	<u>7.207</u>	74.03	<u>94.44</u>	16.13	<u>13.06</u>	47.04	74.27	32.90
ConvONet [167]	Unori	28.30	22.37	77.92	61.76	34.86	20.97	77.80	84.81	35.74	20.06	77.36	87.77	36.04	21.06	77.81	86.33	42.96	8.900	77.30	92.71
IGR [52]	Unori	14.15	66.56	88.28	54.24	12.14	78.23	93.08	45.30	11.42	79.66	93.66	44.26	13.25	74.93	91.72	55.08	28.54	43.63	79.73	73.76
SAP (Gen) [168]	Unori	8.940	57.40	88.72	12.97	8.941	57.48	88.69	13.17	8.922	57.43	88.73	13.04	9.084	56.51	88.42	13.25	10.84	46.57	<u>85.76</u>	16.86
POCO [12]	Unori	8.817	59.91	89.49	<u>13.82</u>	8.809	59.63	89.41	13.75	8.847	59.81	89.44	14.21	8.949	58.87	89.25	14.18	10.44	<u>47.96</u>	86.92	15.37
Neural-Pull [134]	Unori	9.705	69.11	89.12	23.72	6.651	<u>86.51</u>	96.17	15.80	5.674	<u>87.87</u>	<u>96.83</u>	11.11	6.073	81.22	<u>92.69</u>	<u>8.089</u>	10.08	42.95	75.06	12.71
DiGS [8]	Unori	<u>8.559</u>	<u>70.12</u>	<u>90.15</u>	17.54	6.480	85.72	<u>96.45</u>	14.81	6.462	85.79	96.70	15.87	7.654	74.65	91.42	16.86	14.40	26.11	57.33	24.10
NSH [225]	Unori	10.57	64.67	88.78	28.52	8.995	70.75	92.73	21.43	10.26	69.11	91.48	25.19	12.83	50.03	82.58	28.95	17.30	24.13	77.91	31.31
SEik [245]	Unori	9.039	66.46	88.41	17.24	6.134	84.07	95.96	<u>13.09</u>	6.044	85.59	96.75	10.22	7.606	72.61	88.96	12.47	13.16	29.57	57.59	23.29
PG-SDF [84]	Unori	19.27	29.89	74.79	36.89	<u>5.577</u>	83.81	94.30	7.027	<u>5.101</u>	86.44	95.33	<u>6.162</u>	6.414	76.84	90.07	7.463	8.778	53.81	82.18	10.57
I-Filtering [97]	Unori	7.206	77.14	92.54	16.12	<u>5.905</u>	87.43	96.79	13.72	4.792	88.83	97.56	5.653	<u>6.339</u>	<u>80.85</u>	92.81	11.18	<u>9.761</u>	44.90	75.74	<u>12.41</u>
NDF [31]	Unori	11.15	50.18	86.42	26.58	8.840	61.56	93.34	20.98	9.160	59.37	93.66	22.30	10.87	42.72	91.68	23.06	22.09	3.810	87.22	21.76
GIFS [251]	Unori	9.724	56.23	85.45	17.28	7.289	68.08	93.48	9.981	8.183	56.89	93.89	10.82	11.05	28.58	91.34	13.01	22.10	4.640	83.07	22.09
GeoUDF [177]	Unori	6.709	74.51	<u>89.74</u>	9.657	<u>4.855</u>	87.62	96.02	5.130	<u>4.736</u>	88.44	96.80	<u>5.000</u>	8.119	53.88	89.50	8.547	18.39	7.510	79.31	19.39
LevelSetUDF [266]	Unori	8.569	<u>66.64</u>	87.03	15.89	4.895	<u>88.11</u>	<u>96.51</u>	6.205	4.785	<u>88.52</u>	<u>96.90</u>	5.242	<u>6.312</u>	<u>76.10</u>	<u>92.23</u>	6.544	13.46	21.56	83.44	15.87
CAP-UDF [265]	Unori	11.66	62.04	84.36	23.95	4.995	87.67	96.27	6.076	4.917	87.98	96.63	5.853	6.337	75.76	90.91	<u>6.590</u>	<u>11.91</u>	<u>29.72</u>	73.34	<u>15.11</u>
SALS [179]	Unori	<u>8.408</u>	65.69	77.43	<u>14.93</u>	5.123	85.71	92.81	6.362	4.807	87.90	95.17	5.180	7.507	63.13	72.61	9.608	12.44	28.74	58.70	17.12
S ² DF [242]	Unori	14.93	41.52	89.88	32.04	4.776	88.34	97.09	<u>5.372</u>	4.642	89.22	97.82	4.862	6.311	76.67	92.94	6.741	9.326	44.44	86.40	10.73

7 CHALLENGES, EMERGING DIRECTIONS, AND FUTURE OUTLOOK

Taken together, the preceding sections show that implicit geometry is best understood not as a collection of isolated field definitions, but as a coupled pipeline in which representation, extraction, supervision, and application constraints continuously shape one another. Our survey of occupancy, signed distance, unsigned distance, and related implicit fields clarified the distinct trade-offs each formulation makes in regularity, topology, geometric queries, and downstream compatibility, while the benchmark translated those trade-offs into empirical evidence. These observations suggest that the central problems ahead are no longer simply which field to use, but how to design implicit representations and extraction pipelines that are simultaneously more robust, topology-aware, data-efficient, scalable, and controllable. With this perspective in place, we now turn to the major challenges that limit current methods and to the research directions most likely to shape the next stage of progress.

7.1 Non-manifold Structures

Many geometric objects of practical interest are intrinsically non-manifold: material interfaces may meet along shared curves or higher-order junctions, medial axes are inherently sheet-based structures with merges and terminations, and man-made objects often contain thin plates, nested shells, internal partitions, and multi-layer components that create non-manifold configurations.

In these settings, a globally consistent inside/outside labeling becomes ambiguous or ill-posed, making classical signed representations and orientation-dependent pipelines brittle. This is precisely why unsigned and medial implicit representations are increasingly important. Unlike SDFs, UDFs remain meaningful when sign information is undefined, and they naturally accommodate open surfaces, thin structures, and complex topology.

UDF-based reconstruction has already shown strong performance on open and multi-layer geometry, including cases with non-manifold configurations. The main challenge, however, lies not only in representation but also in extraction. For sign-ambiguous fields, standard zero-set extraction methods such as Marching Cubes or Dual Contouring cannot reliably recover consistent sheet connectivity or orientation near intersections, mergers, and branch junctions. This limitation

has been widely observed in recent work: classical methods either fail to detect valid surface elements because no sign changes are present, or they introduce spurious topology when local decisions are ambiguous. Optimization-based methods such as DCUDF [62] and DCUDF2 [23] are robust and can produce a double-layered mesh around the surface, including around non-manifold junctions. However, converting this two-layer membrane into a single non-manifold mesh remains challenging, because front/back separation breaks down at Y-shaped and higher-order junctions, as noted in [86].

Recent efforts have begun to address this challenge from different perspectives. DualMesh-UDF [258] adopts a dual-contouring-style extraction on an adaptive octree. It estimates local tangent planes near the target zero set, solves a least-squares/QEF problem to place one representative surface point in each active cell, and then connects these points into a mesh. Because it does not rely on pseudo-sign estimation, it can in principle recover non-manifold configurations; however, its cell-wise connectivity is fragile, and ambiguous cells often lead to holes or other topological artifacts. MIND [24] instead converts the UDF into a global multi-labeled field that separates different local surface sides and then extracts the resulting material interface with multi-label Marching Cubes. This makes it better suited to orientable open/non-manifold structures than standard MC-based sign-inference methods. However, MIND cannot handle non-orientable surfaces, and because its final extraction stage is still performed on regular cubes via M3C, the number of locally separable sheets is constrained by the cube-based discretization, making very high-valence junctions difficult to recover robustly in practice. To move beyond this grid-induced bottleneck, VoroUDF [218] replaces regular-grid iso-surface extraction with a Voronoi-based optimization. It optimizes surface seeds using an L_1 tangent energy together with a feature-aware repulsive term, and derives mesh connectivity from approximate geodesic Voronoi adjacency, which naturally accommodates more complex non-manifold configurations while also supporting sharp features and open boundaries. DCx [5] extends standard Dual Contouring, which was originally designed for iso-surface extraction from signed distance functions and places one vertex in each active cube detected by sign changes. To adapt DC to UDFs, DCx replaces sign-based active-cube detection with an optimization-guided strategy over each $2 \times 2 \times 2$ voxel block, termed an expanded cube, and introduces a voxel-to-mesh lookup table that encodes valid local connectivity patterns. This redesign enables efficient template-based triangle extraction and supports non-manifold connectivity inside expanded cubes.

Despite recent progress in extracting iso-surfaces with non-manifold structures, several important directions remain open. First, current extractors still provide only limited support for arbitrary junction complexity. Grid-based methods encode a restricted family of local configurations, whereas Voronoi-based methods improve flexibility but still lack formal topological guarantees. Developing topology-complete extractors that can robustly recover high-valence junctions, non-manifold points, and eventually non-orientable structures remains an important open problem. Second, field learning and surface extraction are still largely decoupled. Because UDF errors are typically largest near the zero set, the medial axis, and complex junctions, future methods should jointly learn the field together with side labels, seeds, partitions, or local adjacency, so that topology is enforced during representation learning rather than repaired afterward. Third, robustness to thin layers, closely spaced sheets, and noisy gradients remains limited. This points to the need for adaptive resolution, uncertainty-aware extraction, and explicit topological validation or correction.

7.2 Hybrid Representations and Joint Optimization

Hybrid representations combine complementary field types so that geometric structures that are difficult to recover with any single representation, especially open surfaces, thin sheets, and medial axes, can still be extracted through standard iso-surface pipelines.

A common principle is to decouple metric proximity from topological validity: one component models distance in a way that remains meaningful across the domain, while another determines where sign, phase, or surface existence should be trusted. This idea appears in several forms for surface reconstruction. Points2Surf [42] is an early precursor to hybrid metric–sign decomposition. It factorizes the SDF into an absolute-distance term and a sign component, predicts the absolute distance from local point-cloud patches, estimates the sign from coarse global context, and reconstructs the final SDF as their product before applying Marching Cubes. Hybrid Sign and Distance Field (HSDF) [217] extends this decomposition to arbitrary-topology surface reconstruction. It learns a UDF together with an auxiliary sign field, uses the UDF to localize candidate surface points, and then applies an adaptive masked MC procedure that instantiates a mesh only in surface-supported regions. In this way, HSDF combines UDF-style topological flexibility with SDF-style field-to-mesh conversion, enabling both open and closed surfaces to be reconstructed in a unified framework. SSDF [33] combines a generalized winding-number signal with an unsigned distance field, yielding a semi-signed distance representation that retains proximity information while representing holes and open boundaries implicitly. Three-pole Signed Distance Function (3PSDF) [22] takes a different route by extending the conventional inside/outside signed distance formulation with a third, directionless NULL sign. By assigning null values to regions that contain no target surface, it suppresses unwanted decision boundaries and prevents the implicit field from closing open structures. To avoid the gradient difficulties caused by null-valued targets, 3PSDF formulates learning as a three-way classification problem over {inside, outside, null}, while retaining compatibility with Marching Cubes for mesh extraction. Neural rendering with Arbitrary Topologies (NeAT) [144] brings the same idea into the multi-view neural rendering setting. It represents geometry with an SDF augmented by a differentiable validity branch that predicts the probability that a surface should exist at a queried 3D point. During rendering, low-validity samples are suppressed, and at extraction time low-validity regions are masked out before Marching Cubes by setting their SDF values to NaN. This allows NeAT to reconstruct open as well as watertight surfaces while preserving the rendering and meshing advantages of signed implicit fields. Metric–Phase Fields (MPF) [85] decouple geometry and topology more explicitly by learning an unsigned metric field together with a smooth phase field; the two are fused into a signed implicit function through a gated residual formulation, allowing the representation to behave like an SDF where sign is meaningful while remaining UDF-like in sign-ambiguous thin structures and boundary regions. Unlike Points2Surf [42] and HSDF [217], which decode both sign and distance from a shared feature space, MPF learns them separately and introduces a soft constraint to encourage their alignment along the gradient direction. Such alignment improves numerical stability near the surface, where small inconsistencies can otherwise lead to artifacts.

Hybrid representations are also closely related to joint optimization. For instance, GIFS [251] learns a segment field jointly with a UDF under a shared backbone. The segment field classifies whether two query points are separated by a surface, providing the region-partitioning cues that a standalone UDF lacks for Marching Cubes extraction; the UDF in turn supplies metric proximity for localizing intersection points. In room-level reconstruction, Occ-SDF Hybrid [133] and H2O-SDF [165] each extend a volume-rendered SDF framework with a second field to recover thin indoor structures. The former jointly trains an OF whose per-point occupancy is unaffected by other objects along the ray, compensating for the density distortion that SDF incurs on multi-object rays; the latter introduces an Object Surface Field, a learned surface-probability field whose cross-guidance loss creates non-vanishing gradients with respect to SDF values, enabling convergence at thin structures that would otherwise be lost to vanishing gradients. Q-MDF [86] jointly learns an SDF and a medial field [174], and defines a quasi-medial distance whose zero level set recovers the medial axis.

Across these hybrid representations and joint optimization methods, the key advantage is that iso-surface extraction is applied only where the auxiliary sign, validity, or phase information is meaningful, which suppresses the unwanted closures and topological distortions that arise when a single signed field is forced to explain open or sign-ambiguous geometry.

Several future directions appear particularly important for hybrid representations. First, representation and extraction remain largely decoupled: the field is learned first, and topology is recovered afterward using Marching Cubes variants or post-processing. A promising direction is therefore to co-design hybrid fields with topology-aware extraction structures, such as material-interface partitions [24], adaptive cell complexes [5], or Voronoi-based discretizations [218], so that topology is enforced during learning rather than repaired afterward.

Second, the coupling between geometric and topological components is often still weak. Most existing methods combine metric and sign/validity cues in a loosely connected manner, which can lead to inconsistencies near boundaries, thin structures, and junctions. This suggests the need for tighter analytical and optimization-level coupling, including uncertainty-aware phase prediction, adaptive transition sharpness, and explicit consistency constraints near boundaries and high-order junctions.

Third, since hybrid representations are typically designed to address challenging scenarios where conventional implicit functions fail, evaluation protocols should move beyond purely geometric errors. Metrics such as Chamfer distance or Hausdorff distance are insufficient to capture the intended improvements. More informative evaluation should explicitly measure boundary fidelity, thin-sheet separation, phase/sign consistency, and junction correctness, particularly for non-manifold configurations.

7.3 Spatial Data Structures

Spatial data structures do not merely accelerate implicit methods; they also define the neighborhoods over which fields are optimized, regularized, and extracted. In practice, different structures induce different notions of locality and therefore different trade-offs between regularity, adaptivity, and geometric fidelity. Octrees provide hierarchical volumetric support; sparse voxels and adaptive grids retain an explicit discretization while concentrating memory and resolution near occupied, high-curvature, or topologically ambiguous regions; point sets and natural neighbors follow the sample distribution without committing to a fixed voxelization; Voronoi decompositions replace axis-aligned cells with geometry-aware partitions; and multiresolution hash tables act as compressed learned lattices for neural fields. From this perspective, the choice of spatial data structure is not merely an implementation detail, but a modeling decision that determines where representational capacity is allocated and which geometric or topological features are easiest to preserve.

Octrees remain the most commonly used spatial scaffold for implicit reconstruction. In the Poisson Surface Reconstruction family, Kazhdan et al. [78] use an adaptive octree to represent the implicit function and solve the Poisson system only where detail is needed; Screened Poisson Surface Reconstruction [79] retains the same hierarchy while adding interpolation constraints; and iPSR [61] extends the framework to unoriented points by iteratively updating normals from the reconstructed surface. The same logic persists in recent learned-UDF meshing: DualMesh-UDF [258] uses an adaptive octree to localize tangent-plane estimation and surface-point placement.

Adaptive grids and sparse voxel structures pursue a similar principle, but with a more explicit notion of occupancy. Rather than storing features over a dense lattice, methods such as NSVF [108] organize voxel-bounded implicit fields in a set of sparse voxels, allowing computation to skip empty space and focus on occupied regions. Voxurf [231] follows the same philosophy for neural surface reconstruction: it combines an explicit voxel-based SDF representation with a two-stage coarse-to-fine optimization procedure and hierarchical geometry features that encourage information sharing

across voxels, thereby improving both efficiency and surface fidelity. Taken together, sparse voxels and adaptive grids can be understood as generalizations of the octree philosophy: they couple implicit functions to discretizations that place representational capacity where geometry is present, intricate, or topologically ambiguous.

Voronoi-based structures offer a complementary, sample-aware alternative to voxel hierarchies. In GCNO [239], Voronoi vertices and outside Voronoi poles are used to regularize the winding-number field and recover globally consistent normal orientations from raw point clouds; BIM [118] is a closely related follow-up in the same winding-number line that replaces this with a boundary-integration energy; and VAD [87] uses Voronoi-based geometric criteria to assign bidirectional normals before diffusing them into an approximate UDF gradient field. At the meshing stage, VoroUDF [218] goes further and replaces fixed grid cells with Voronoi cells, ultimately extracting connectivity from an approximate geodesic Voronoi diagram on the implicit zero set. Seen in this light, recent Voronoi-based neural/UDF methods reconnect with older Voronoi-interface work in multiphase implicit geometry, where Voronoi partitions were already used to track and mesh interconnected implicit interfaces.

Point-based structures are equally important, especially when the input is irregular, sparse, or thin. Point Set Surfaces [1] treats the point cloud itself as the primary geometric carrier and defines a smooth surface via local moving-least-squares projections. VIPSS [71] reformulates point-based surface reconstruction as a variational implicit problem on unoriented points, showing that the field can be defined directly from points rather than from a volumetric grid. NN-VIPSS [232] makes this idea substantially more scalable by exploiting the locality of natural neighborhoods, effectively turning a global variational formulation into a locally supported one without sacrificing robustness on sparse and non-uniform inputs. This point- and natural-neighbor-based perspective is particularly attractive for thin structures, where fixed voxel neighborhoods can be too rigid and geometry-aware local support is often more reliable.

Finally, multiresolution hash tables have become a dominant learned spatial structure for neural implicit fields. Instant-NGP [151] introduces the multiresolution hash table as a compact trainable encoding for coordinate networks. Neuralangelo [100] shows how multi-resolution 3D hash grids, combined with coarse-to-fine optimization, can drive high-fidelity neural surface reconstruction. Although these encodings are not meshing structures in the classical sense, they play a closely related role: they determine where representational capacity is concentrated in space, and thus strongly influence both optimization and the fidelity of the learned implicit function.

Overall, spatial data structures serve not only as acceleration mechanisms, but also as structural priors on locality, resolution, and topology.

7.4 New Neural Models

Recent work has increasingly moved beyond the standard formulation of neural implicit fields as a single smooth coordinate-based MLP representing an occupancy function, SDF, or UDF. These new neural models can be understood from two perspectives: representation power and downstream usability. The former concerns how network architectures can better encode geometric, topological, or analytic structures that are difficult for generic coordinate MLPs to learn, while the latter concerns how neural implicit fields support practical downstream use beyond pure representation quality.

From the representation perspective, recent neural models increasingly focus on sharp features, discontinuities, or other forms of nonsmooth behavior. Standard coordinate-based MLPs are biased toward smooth functions, so discontinuities, creases, and jump-like behavior are often approximated by rapidly varying continuous transitions. Recent models instead introduce explicit mechanisms for nonsmoothness, such as duplicating degrees of freedom across discontinuity sets, learning the

locations of such sets, or lifting the input space with auxiliary fields. In this way, sharp features become part of the representational model rather than artifacts that must be recovered after training.

Discontinuity-aware neural fields [7, 107] are designed to represent functions with localized discontinuities, such as image edges. Belhe et al. [7] use a fixed triangulation and assign separate feature values to the two sides of each triangle edge. Barycentric interpolation within the triangulation then induces discontinuities across edges. Liu et al. [107] extend this idea by making the discontinuity boundaries learnable, allowing the model to adapt the discontinuity set to the data and capture more flexible structured discontinuities. Chang et al. [18] use the generalized winding number as an auxiliary spatial lifting for discontinuity modeling. Since the generalized winding number can be interpreted as the solution of a Poisson problem with a jump Neumann boundary condition, it provides a physically motivated lifting coordinate that helps the network capture discontinuous, or C^{-1} , behavior. These approaches are primarily designed for 2D and 2.5D data, where discontinuity curves can be represented and optimized more directly. SharpNet [160] is an MLP variant for representing functions that are smooth almost everywhere while exhibiting non-differentiability at prescribed locations. It augments the network input with a learnable auxiliary feature function that encodes the desired sharp-feature set. By exposing this feature information to the MLP, SharpNet enables localized nonsmooth behavior without forcing the entire network to approximate sharp features through high-frequency variation. SharpNet can be used to encode both signed and unsigned distance fields for representing 2D and 3D geometry. It is particularly suitable for applications where non-differentiability arises naturally, such as geodesic distance fields, medial fields, and CAD models, in which creases, ridges, and junctions introduce intrinsic nonsmooth structures. This is consistent with the observation that distance-based representations often exhibit non-differentiability at the surface, medial axis, or other singular sets, which are challenging for standard neural fields to approximate accurately.

Beyond representation power, downstream usability raises a different set of requirements. Three directions are particularly important: local controllability for direct modeling and editing, scalable large-scene representations, and efficient field queries for downstream geometric computation.

First, local controllability remains a major limitation of current neural implicit representations. Most existing models are globally parameterized, encoding geometry in a highly entangled manner across the entire domain, so local modifications may induce unintended global changes. A key direction is therefore to develop representations with stronger locality, where geometric variations can be confined to restricted spatial regions without affecting the rest of the shape. This may be achieved through localized basis functions, spatial partitioning, or hierarchical architectures that decouple global structure from local details. Stronger locality also provides a foundation for geometric editing. Unlike explicit representations, which provide direct handles through vertices, edges, faces, or control structures, implicit fields do not naturally expose intuitive operations for selecting, moving, or deforming local parts of a shape. Future work could address this gap by establishing bidirectional mappings between explicit geometric operations and implicit representations, enabling controllable interactive editing while preserving geometric consistency. Beyond editing, most neural implicit representations remain reconstruction-oriented, learned or fitted from external observations such as point clouds, meshes, or multi-view images rather than designed as user-facing modeling paradigms. A further direction is therefore to develop implicit representations as media for direct geometric design, with more interpretable parameterizations and interactive mechanisms that translate high-level operations into controlled updates of the underlying field. Overall, building neural implicit models that are simultaneously expressive, locally controllable, and amenable to interactive manipulation remains a significant open challenge, and a necessary step toward enabling implicit representations to serve as a universal geometric substrate rather than merely as intermediate representations.

Second, scalability requires locality at the level of scene organization. A single global MLP is often inadequate for large objects or scene-level geometry, because its capacity is shared across the entire domain and distant regions are coupled through the same parameters. Recent work therefore treats scalability as a problem of spatial decomposition: instead of representing the whole scene with one global field, the model is organized into local implicit functions with controlled support, resolution, and coupling. Regular or overlapping local grids [75, 16], adaptive octree decompositions [211], and graph-based compositions of overlapping local SDFs [246] all follow this principle in different forms. The key challenge is not merely to increase model capacity, but to allocate it locally and adaptively while maintaining consistency across neighboring fields, so that large scenes can be represented without sacrificing fine geometric detail.

Third, efficient field queries are essential for downstream geometric computation. Meshing, rendering, collision detection, simulation, and path planning require repeated evaluations of field values, gradients, normals, or intersections. Recent architectures suggest that efficiency can be built into the implicit representation itself rather than handled only by external acceleration structures. A representative line is T-MLP [244], which attaches output tails to intermediate layers so that valid predictions can be obtained at different network depths. SAND [243] extends this idea by coupling the tailed network with a spatial network-depth map, allowing each query to be evaluated only to the depth required by its spatial importance and local geometric complexity. This turns neural field evaluation into an adaptive process and naturally supports level-of-detail control. Future work may further explore learned depth prediction, training-time adaptive computation, and redundancy-aware pruning, enabling more controlled trade-offs among accuracy, runtime, and memory.

7.5 Implicit Representation-based 3D Geometry Compression

Implicit geometry compression converts a mesh into an implicit field, compresses the field, and reconstructs the surface via iso-surface extraction, bypassing the need to code irregular mesh connectivity directly.

On dense grids, Tang et al. [210, 209] partition the TSDF into local blocks for real-time streaming of dynamic performance sequences, progressing from a linear codec to an end-to-end auto-encoder with a learned entropy model and lossless sign coding to preserve topology. NeCGS [178] targets geometry-set compression: each mesh is converted into a volume combining TSDF with per-vertex displacement vectors, and a shared auto-decoder amortizes inter-shape redundancy. SIR-SNC [216] moves to the sparse setting, retaining field values only on near-surface grid vertices and compressing the resulting sparse tensor with a sparse-convolutional auto-encoder.

Several future directions can substantially advance this area. Current methods operate almost exclusively on (T)SDF over regular or sparse grids; extending implicit compression to unsigned or hybrid representations would enable the coding of open surfaces, thin sheets, and non-manifold geometry that SDFs cannot faithfully encode. Continuous implicit fields, with their resolution-independent expressiveness, are also a natural fit for compression. Investigating compact architectures specifically optimized for this purpose remains largely unexplored, particularly in offline scenarios where higher encoding cost is acceptable. More fundamentally, existing pipelines first fit the implicit field and then independently compress it, yet jointly optimizing the representation together with compression could yield substantially better rate-distortion trade-offs. Finally, practical assets rarely consist of geometry alone, and jointly compressing geometry with surface attributes such as color and material parameters within a unified implicit framework is an important open problem that could build on recent advances in neural texture compression [46].

8 CONCLUSION

In this paper, we have presented a comprehensive survey of implicit geometric representations, their surface extraction algorithms, and their downstream applications from a representation-centric perspective. We unified diverse implicit representations as a pair $\mathcal{F} = (f : Q \rightarrow \mathcal{V}, C)$ of a field function and a geometry condition. Under this formulation, we reviewed occupancy, signed distance, and unsigned distance fields, together with their vector, ray, segment, medial, and hybrid extensions. Organized along field types, we further surveyed iso-surface extraction algorithms, from classical Marching Cubes and Dual Contouring to recent UDF-tailored variants. Building on this backbone, we then revisited two downstream applications, namely surface reconstruction and 3D geometry generation. Within each application, we traced the evolution of methods along the axis of implicit representations. To assess the practical behavior of existing methods, we contributed a benchmark on point cloud surface reconstruction, the longest-standing application of implicit fields, covering watertight objects, non-watertight objects, synthetic scenes, and real scans. Under a unified protocol, the benchmark evaluates 19 representative classical and neural field methods. Finally, we discussed the major limitations of current implicit geometry methods and outlined several promising directions for future research. We hope this survey provides a useful reference for understanding the trade-offs of implicit representations and for designing methods that better align representation choice with algorithmic requirements and application goals.

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